Semantics of Split Quantification

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Contents

1 Introduction

This paper investigates the semantics of split quantification, in which quantificational elements appear morphosyntactically separated from the nominal expressions they are interpretively associated with. The empirical focus is on three phenomena that instantiate split quantification: floating quantifiers, quantification at a distance, and reverse proportional measure phrases. These phenomena exhibit systematic interpretive effects that distinguish them from their non-split counterparts, raising fundamental questions about the compositional mechanisms underlying split quantification and the relationship between morphosyntactic form and semantic interpretation.

The defining characteristic of split quantification is that quantifiers are separated from their associated nominal material, rather than being contained within a single nominal projection. Consider the following minimal pair illustrating quantification at a distance (QAD) in French. The quantifier *beaucoup* 'a lot' can appear either within the nominal projection, as in (1a), or separated from the *de*-phrase it typically combines with, as in (1b), yielding corresponding interpretive differences (Obenauer 1984):

(1) a. Le maire a salué [**beaucoup** de sportifs]
the mayor has greeted a.lot of sportsmen

'The mayor greeted many sportsmen.' (✓collective; ✓distributive)

b. Le maire a **beaucoup** salué de sportifs
the mayor has a.lot greeted of sportsmen

'The mayor greeted many sportsmen.' (✗collective; ✓distributive)

While both sentences are felicitous in contexts where the mayor greets many sportsmen, only (1a) can describe a single collective greeting event involving a large group. The split configuration in (1b) requires multiple individual greeting events, showing that the structural separation between *beaucoup* 'a lot' and *de sportifs* 'of sportsmen' systematically affects interpretation.

Similar split patterns emerge with reverse proportional measure phrases in Korean.

(2) a. Nari-ka [yukinong ceyphum-uy **50%**]-lul sassta
Nari-NOM organic product-GEN 50%-ACC bought

'Nari bought 50% of the organic products.' (✓proportional)

b. Nari-ka [yukinong ceyphum]-ul **50%** sassta
Nari-NOM organic product-ACC 50% bought

'50% of what Nari bought was organic products.' (✓reverse proportional)

In (2a), the measure phrase *50%* appears with a partitive-marked noun phrase and receives a straightforward proportional interpretation: Nari bought half of the contextually salient organic products. By contrast, in (2b), the measure phrase is separated from the noun by accusative case. Crucially, the interpretation reverses: half of what Nari bought was organic products. As observed by Ahn and Sauerland (2017), this reversal presents an apparent challenge to the theoretical constraint that every quantificational determiner is conservative, since the proportion appears to be calculated relative to the nuclear scope (buying events) rather than the restrictor (organic products).

More broadly, both quantification at a distance and reverse proportional measure phrases exhibit event-sensitive restrictions that are absent in their non-split counterparts. These restrictions closely parallel those observed with another instance of split quantification, namely floating quantifiers in Japanese (Nakanishi 2007). The interpretive contrast between split and non-split configurations calls for an explicit compositional semantic account that applies across these split phenomena.

The central claim advanced in this paper is that split quantification involves quantification over events rather than individuals, with two core compositional ingredients. First, split quantifiers are polymorphic: they denote the same function based on part-whole structure, applying uniformly to properties of individuals (type $\langle et \rangle$) or properties of events (type $\langle vt \rangle$). Second, the structural position of NP arguments relative to thematic predicates correlates with each quantification: when the NP combines with the quantifier before the thematic predicate applies, individual quantification results; when the split quantifier applies after the NP combines with the thematic predicate, event quantification emerges. Together, polymorphism and the scope of thematic saturation provide the formal apparatus for encoding split quantification in natural language. The analysis is formulated within a neo-Davidsonian framework in which arguments compose via modification and are linked to verbal predicates through thematic relations. This event-quantificational approach accounts for the observed event-sensitive restrictions in a principled manner.

The organization of the paper is as follows. Section 2 establishes the empirical landscape by introducing split quantification as a morphosyntactic phenomenon with systematic interpretive effects. Three phenomena are presented: floating quantifiers, quantification at a distance, and reverse proportional measure phrases. Floating quantifiers in Japanese serve as a starting point, establishing the event-sensitive restrictions that motivate an event-based approach. The section also distinguishes the type of event-sensitivity at issue in split quantification from the well-known phenomenon of event-relatedness in the nominal domain, as in *4,000 ships passed through the lock last year* (Krifka 1990).

Section 3 examines quantification at a distance (QAD) in French, starting with previous observations that QAD *beaucoup* 'a lot' imposes both a multiplicity of events and a multiplicity of objects. I argue that while the event-side generalization indeed involves quantification over events as a semantic requirement, the object-side requirement is not semantically encoded but rather arises as a cancelable and context-dependent pragmatic inference. The analysis assigns *beaucoup* 'a lot' a uniform semantics across its non-split and split configurations, with variation only in the type of argument it combines with. The compositional semantics to be proposed also derives the interpretive contrast between the split and non-split forms in a way that is compatible with both types of syntactic analyses of QAD previously proposed.

Section 4 presents a compositional analysis of reverse proportional measure phrases, focusing primarily on Korean, a language that allows reverse proportional interpretations with a range of argument types. I show that measure phrases in split configurations can associate with agents, themes, goals, and temporal and locative modifiers, and can also occur in attributive and small-clause contexts. I provide explicit derivations demonstrating how the proposed semantics applies across these different structural environments. The section examines event-sensitive restrictions, showing how they follow from the event-quantificational analysis, and addresses the interpretive ambiguity unique to Korean split quantification through an account based on multiple case constructions. As with quantification at a distance, the proposal relies on polymorphism over individuals and events, though the event-based account of split quantifiers is here augmented to capture reverse proportionality in measurement sentences. Section 5 concludes.

2 Introducing split quantification

This section introduces split quantification, a morphosyntactic phenomenon in which quantificational elements are distributed across multiple positions within a clause rather than being contained within a single nominal projection. Section 2.1 presents floating quantifiers, establishing the event-sensitive restrictions that motivate an event-based approach to split quantification. Section 2.2 introduces quantification at a distance in French, where the quantifier *beaucoup* 'a lot' can appear separated from its associated nominal phrase. Section 2.3 turns to reverse proportional measure phrases: expressions like *50%* receive interpretations in split configurations that reverse the standard proportional reading. Section 2.4 distinguishes the type of event-sensitivity at issue in split quantification from event-relatedness in the nominal domain, clarifying the similarities and differences between these two phenomena. This discussion situates the event-based analysis presented in subsequent sections within the broader landscape of event-relatedness and quantification.

2.1 Floating quantifier

A canonical example of split quantification is the so-called floating quantifier construction, where a quantifier surfaces away from the nominal it is interpretively associated with. For instance, (3) shows that both the determiner position and the adverbial position may host an overt quantificational element in French (Sportiche 1988).

(3)

{Tous}	les	enfants	ont	{tous}	vu	ce	film
all	the	children	have	all	seen	this	movie

'All the children have seen this movie.'

The question that has extensively been discussed in the literature is whether the floating quantifier originates in the DP and is stranded by movement, or whether it is base-generated as an adverbial element in the clause. On the stranding analysis (Sportiche 1988; Shlonsky 1991; Merchant 1996; McCloskey 2000, among many others), the floating quantifier forms a constituent with its associate at some stage of the derivation. Subsequent movement of the associated DP leaves the quantifier stranded, producing the observed surface order. On the adverbial analysis (Doetjes 1992; Baltin 1995; Torrego 1996; Bobaljik et al. 2003, among many others), the floating quantifier does not form a constituent with the DP at any derivational stage. Instead, it is base-generated as an adverbial adjunct modifying the verb phrase. Despite their surface differences, the two strings in (3) receive the same interpretation, which does not provide conclusive evidence in favor of one syntactic analysis over the other. Since the meaning of the sentence remains identical regardless of their position, floating quantifiers like the one in (3) have not been the focus of much investigation on the semantic side.

However, there are floating quantifier constructions that have been shown to result in different interpretive effects. Nakanishi (2007) shows that numerals and measure phrases in Japanese, such as *san-nin* 'three-CL' and *san-rittoru* 'three liter,' can occur in the so-called split constructions, as shown in (4b).

(4) a.

[gakusei	**san-nin**]-ga	ie-ni	kaet-ta
student	three-CL-NOM	home-to	go-PAST

b.

gakusei-ka	ie-ni	**san-nin**	kaet-ta
student-NOM	home-to	three-CL	go-PAST

'Three students went home.'

Constructions like (4b) are split in the sense that the noun phrase and the numeral are clearly separated by a case marker and/or an additional argument. Nakanishi shows that numerals and measure phrases in such configurations exhibit three interpretive restrictions not found in (4a)-type configurations. Specifically, numerals and measure phrases in split configurations are incompatible with (i) collective predicates, (ii) individual-level predicates, and (iii) single-occurrence events. The examples below, from Nakanishi (2007), illustrate each restriction.

(5) a.

[tomodati	**huta-ri**]-ga	kyonen	kekkonsita
friend	two-CL-NOM	last.year	married

'Two friends got married last year.' (distributive ✓; collective ✓)

b.

tomodati-ga	kyonen	**huta-ri**	kekkonsita
student-NOM	last.year	two-CL	married

'Two friends got married last year.' (distributive ✓; collective ✗)

(6) a. [kaba **san-too**]-ga genkidearu
hippo three-CL-NOM healthy
'Three hippos are healthy.' (✓stage-level)

b. kaba-ga **san-too** genkidearu
hippo-NOM three-CL healthy
'Three hippos are healthy.' (✓stage-level)

c. [kaba **san-too**]-ga osudearu
hippo three-CL-NOM male
'Three hippos are male.' (✓individual-level)

d. ??kaba-ga **san-too** osudearu
hippo-NOM three-CL male
'Three hippos are male.' (✗individual-level)

(7) a. [gakusei **san-nin**]-ga kinoo sono isu-o kowasita
student three-CL-NOM yesterday that chair-ACC broke

b. ??gakusei-ka kinoo **san-nin** sono isu-o kowasita
student-NOM yesterday three-CL that chair-ACC broke
'Three students broke that chair yesterday.'

Based on these event-sensitive restrictions, Nakanishi argues that floating quantifiers such as *san-nin* 'three-CL' are modifiers in the event domain. In Sections 3 and 4, I show that similar event-sensitive restrictions arise in French quantification at a distance and in Korean reverse proportional measure phrases, motivating event-based compositional analyses of both phenomena.

2.2 Quantification at a distance

The second example of split quantification is quantification at a distance (QAD), well documented in French. The quantifier *beaucoup* 'a lot' may appear separated from the nominal it typically combines with, as shown in (8b) (Doetjes 1997).

(8) a. Jean a lu [**beaucoup** de livres]
Jean has read a.lot of books

b. Jean a **beaucoup** lu de livres
Jean a a.lot read of books
'Jean read a lot of books.'

Similar to floating quantifiers, QAD has often been discussed concerning whether *beaucoup* in (8b) undergoes syntactic movement from the nominal position (e.g., Boivin 1999) or is base-generated in the surface adverbial position (e.g., Doetjes 1997). However, unlike floating quantifiers like *tous* 'all,' QAD *beaucoup* has interesting interpretive effects, as noted by Obenauer (1984). For instance, (9a) can be true either if the mayor greeted a crowd of sportsmen collectively or if many sportsmen were greeted individually. In contrast, (9b) can only be interpreted in the latter way.

(9) a. Le maire a salué [**beaucoup** de sportifs]
the mayor has greeted a.lot of sportsmen
'The mayor greeted many sportsmen.' (✓collective; ✓distributive)

b. Le maire a **beaucoup** salué de sportifs
the mayor has a.lot greeted of sportsmen
'The mayor greeted many sportsmen.' (✗collective; ✓distributive)

Beaucoup as a split quantifier exhibits other event-sensitive restrictions that parallel those in Japanese floating quantifiers, which are not observed in nominal examples like (9a). In Section 3, building on Burnett (2012), I present a compositional semantic account that captures these empirical generalizations while remaining compatible with both syntactic analyses.

2.3 Reverse proportional measure phrase

The third instance of split quantification is found with measure phrases such as *50%*. Like the floating quantifiers and quantification at a distance discussed earlier, these measure expressions, when split, exhibit event-related contrasts with their non-split counterparts. Such patterns strongly suggest that split quantification fundamentally involves event structure. However, reverse proportional measure phrases introduce an additional interpretive puzzle: the apparent reversal of proportional calculation, illustrated for Korean in (10b) (Ahn and Sauerland 2017).

(10) a. hoysa-ka [yeca-uy **50%**]-lul ceyyonghayssta
company-NOM women-GEN 50%-ACC hired

'The company hired 50% of the women.' (✓proportional)

b. hoysa-ka yeca-lul **50%** ceyyonghayssta
company-NOM women-ACC 50% hired

'50% of the hires were women.' (✓reverse proportional)

In (10a), the measure phrase *50%* appears with a genitive-marked noun phrase and receives a straightforward proportional interpretation. However, in (10b), where the measure phrase is split from the nominal by the accusative case marker, the interpretation reverses: half of the hires were women. This reversal potentially challenges the theoretical constraint that every quantificational determiner is conservative (Barwise and Cooper 1981; Keenan and Stavi 1986), since the proportion appears to be calculated relative to the nuclear scope (hiring events) rather than the restrictor (women). The pattern in (10) also occurs in English, as shown in (11). The split nature is more subtle, however, since it is not reflected in the presence of an overt case marker but in the absence of the partitive marker.

(11) a. The company hired [50% of the women]. (✓proportional)

b. The company hired 50% women. (✓reverse proportional)

Section 4 develops a compositional analysis of reverse proportional measure phrases, treating them as event quantifiers parallel to the QAD quantifiers from Section 3, while also accounting for the reversal effect. The section begins by reviewing prior approaches to reverse proportional readings of quantity adjectives such as *many* and *few*, which have received considerable attention in the literature. These analyses provide essential background for developing an event-quantificational account that also captures reverse proportionality in measurement sentences.

2.4 Event-relatedness across nominal and verbal domains

In the following sections, I show that the split quantification phenomena discussed above involve quantificational expressions that interact fundamentally with event structure. It is therefore important to distinguish the kind of event sensitivity at issue in split quantification from the type of event-relatedness long recognized in the nominal domain. Krifka (1990) famously observed that sentences containing numerals and measure phrases can be ambiguous between object-related and event-related readings, as illustrated in (12).

(12) a. 4,000 ships passed through the lock last year. (✓object-related; ✓event-related)

b. The library lent out 23,000 books in 1987. (✓object-related; ✓event-related)

Under the object-related reading of (12a), there are 4,000 distinct ships involved; each ship is counted once, regardless of how many times it passed. Under the event-related reading, by contrast, the quantifier counts events of passing through the lock, so the same ship may contribute multiple times. The sentence is true if there were 4,000 passing-through events, even if fewer than 4,000 ships exist.

The existence of two readings is also found in Japanese floating quantifiers. Nakanishi (2007) argues that the mechanism responsible for the event-related readings in (12) as well as (13a) must be independent of the mechanism underlying Japanese split quantification in (13b), based on the fact that both the non-split and split numeral sentences in (13a) and (13b) allow the object-related and event-related readings in the sense described above.

(13) a. [fune **yonsen-soo**]-ga kyonen kono suimon-o tuukasi-ta
ship 4000-CL-NOM last.year this lock-ACC pass-PAST

b. fune-ga kyonen **yonsen-soo** kono suimon-o tuukasi-ta
ship-NOM last.year 4000-CL this lock-ACC pass-PAST

'4,000 ships passed through this lock last year.' (✓object-related; ✓event-related)

In Krifka's original work, the flexibility in the nominal domain (12) is captured by event-object mapping principles that allow the quantifier to apply either to the set of individuals or to the set of event tokens associated with them. For concreteness, I follow the analysis of nominal event-relatedness proposed by Musan (1995) and Barker (1999), rooted in Carlson's (1977) notion of stages. On this view, event-related readings arise from individuating entities in terms of relevant stages, corresponding to different temporal slices of individual ships.[1]

Now consider proportional measure phrases in English and Korean.

(14) a. [**50%** of the ships] passed through the lock last year.

b. caknyeney [pay-uy **50%**]-ka sumun-ul cinakassta
last.year ship-GEN 50%-NOM lock-ACC passed

'50% of the ships passed through the lock last year.'

Both English and Korean proportional sentences in (14) allow only the object-related reading, as they measure a proportion of a contextually salient set of ships. However, once a salient basis for individuating ships is provided, as in (15), the event-related reading can arise.

(15) [**50%** of [the ships that loaded cargo at the port]] passed through the lock last year.

(15) can be uttered to mean that 50% of the times that a ship was loaded with cargo, that ship went through the lock. For example, suppose 4,000 ships loaded cargo at the Port of Los Angeles, but they do not correspond to 4,000 distinct ships. Instead, there are fewer than 4,000 ships, some of which passed through multiple times. In this context, where the salient ship stages number 4,000, (15) can still express that 50% of those ship stages that loaded cargo passed through the lock, i.e., an event-related reading.

What remains is whether split measurement sentences, such as (16) in Korean, allow both object-related and event-related readings, and if so, how to account for the ambiguity in split quantification.

(16) pay-ka caknyeney **50%** sumun-ul cinakassta
ship-NOM last.year 50% lock-ACC passed

'50% of what passed through the lock last year were ships.'
(✓object-related reverse proportional; ✓event-related reverse proportional)

[1]Beyond numerals and measure phrases, another class that shows ambiguity between object-related and event-related readings in the nominal domain includes some (but not all) frequency adjectives, such as *occasional*. In the literature on *occasional*, these readings are often characterized as internal (object-related) and adverbial (event-related). The adverbial reading of *occasional* has received (i) a binary quantifier analysis (Zimmermann 2003), (ii) a kind-based analysis (Gehrke and McNally 2015), and (iii) a stage-based analysis (Sant and Ramchand 2022).

In (16), the measure phrase *50%* is split from the nominal, resulting in a reverse proportional reading. Crucially, this sentence supports both object-related and event-related reverse proportional readings. Under the event-related reading, out of 100 total passings, there may be fewer than 50 distinct ships, but multiple crossings per ship can result in 50% of the total passing events being ship events. Under the object-related reverse proportional reading, there were 50 distinct ships, as opposed to, say, yachts, boats, or submarines, among the 100 distinct watercraft that passed through the lock last year.

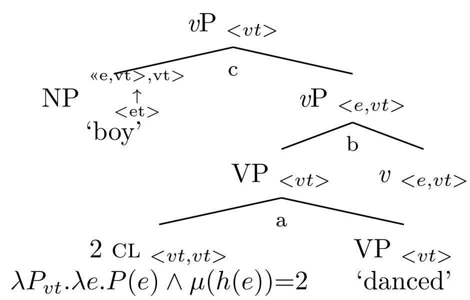
The availability of both readings in split sentences suggests that split quantification introduces an additional dimension to the range of available interpretations. In (16) as well as Japanese split numerals in (14b), the salient domain of quantification is events. Object-relatedness in split quantification can then be analyzed as quantification over event participants. This is precisely the approach taken by Nakanishi (2007) for object-related readings of numerals and absolute measure phrases in Japanese split quantification.

Building on Krifka (1989), who assumes a homomorphism from events to runtimes, Nakanishi proposes a homomorphism h from events to participants. A homomorphism is a structure-preserving function from one domain to another. With this function h, $h(e)$ in (17b) denotes the agents of dancing, and the star operator pluralizes a singular predicate, generating sums of their members in the individual domain (Link 1983) and in the event domain (Landman 1989).

(17) a. otokonoko-ka **huta-ri** odotta
boy-NOM two-CL danced

'Two boys danced.'

b.



$[\![a]\!] = \lambda e. *dance(e) \wedge \mu(h(e)) = 2$
$[\![b]\!] = \lambda x.\lambda e.ag(e) = x \wedge *dance(e) \wedge \mu(h(e)) = 2$
(via EVENT IDENTIFICATION; Kratzer (1996))
$[\![boy]\!] = \lambda x. *boy(x) \rightarrow \lambda P_{e,vt}.\lambda e.\exists x[*boy(x) \wedge P(x)(e)]$
(via A; adapted from Partee (1987))
$[\![c]\!] = \lambda e.\exists x[*boy(x) \wedge ag(e) = x \wedge *dance(e) \wedge \mu(h(e)) = 2]$
$\rightarrow \exists e.\exists x[*boy(x) \wedge ag(e) = x \wedge *dance(e) \wedge \mu(h(e)) = 2]$ (via $\exists_e$)

Nakanishi's compositional semantics in (17b) based on a homomorphism from events to participants clearly targets the object-related reading of split sentences. For example, the Japanese sentence in (18) is felicitous in a context where two students coughed, with one of them coughing multiple times; yet Nakanishi's semantics for the split numeral 'two-CL' ensures that $\mu(h(e)) = 2$. I follow Nakanishi in deriving the object-relatedness of split quantification. By contrast, the compositional account to be developed in the following sections targets event-related readings of split quantification.

(18) gakusei-ka kono jup-pun-de **huta-ri** seki-o si-ta
student-NOM this ten-minute-in two-CL cough-ACC do-PAST

'Two students coughed within the last ten minutes.'

To sum up, object-related and event-related ambiguity is observed in both the nominal domain and its split variant. In nominal, non-split quantification, basic quantification remains in the individual domain. To account for the event-related reading, stages or event tokens provide an alternative way of individuating entities, as proposed by Barker and Musan. In split quantification, by contrast, quantification fundamentally takes place in the event domain. To account for the object-related reading, individuals are accessed through their role as event participants, formalized as a homomorphism, following Nakanishi. The compositional analysis I propose in the following sections will account for event-relatedness in split quantification. Recognizing the two-way distinction—object-related versus event-related in the interpretive dimension, and split versus non-split in the morphosyntactic dimension—is essential for understanding the full empirical picture and for situating event sensitivity in split quantification within the broader discussion of event-relatedness. At the same time, the symmetric pattern between nominal (non-split) and split quantifiers supports an analytical parallel across the two configurations, specifically an analysis of object-relatedness and event-relatedness grounded in the relevant domain of quantification: individuals versus stages in the non-split configuration, and events versus event participants in the split configuration.

3 Quantification at a distance

3.1 Introduction

This section examines Quantification at a Distance (QAD) in French, focusing on the behavior of *beaucoup* 'a lot' as a split quantifier. In QAD configurations, *beaucoup* appears separated from the *de*-phrases it typically occurs next to, giving rise to distinctive interpretive effects absent in non-split configurations. QAD *beaucoup* has received a compositional semantic analysis as an irreducible binary quantifier over events and individuals (Burnett 2012). Based on empirical data not fully addressed in the existing analysis, I argue that while the event-side generalization indeed involves quantification over events, the object-side requirement is not encoded in the semantics of *beaucoup*. The analysis remains compatible with the two major syntactic analyses proposed for French QAD. I also show that the key restrictions on QAD *beaucoup* observed in the literature, namely its incompatibility with collective events, stative predicates, and single-event contexts, follow from the proposed analysis.

The event-quantificational approach developed here builds directly on the core insight that split quantifiers operate in the event domain, as proposed by Nakanishi (2007) for Japanese floating quantifiers. The proposal assigns *beaucoup* a uniform semantics across its non-split and split positions, differing only in the type of arguments it combines with: properties of individuals versus properties of events. As will be shown in Section 4, this analysis extends to another instance of split quantification, reverse proportional measure phrases, which exhibit event-sensitive restrictions while also introducing the puzzle of proportional reversal.

3.1.1 Previous analyses of quantification at a distance

Quantification at a distance (QAD) in French occurs with degree expressions such as *beaucoup* 'a lot,' *peu* 'little,' and *trop* 'too much.' In (19a), *beaucoup* occurs in its usual nominal position, combining directly with a *de*-phrase. By contrast, *beaucoup* in (19b) appears before the verbal element, separated from the *de*-phrase, creating a QAD configuration.[2]

(19) a. Jean a lu [**beaucoup** de livres]
Jean has read a.lot of books

b. Jean a **beaucoup** lu de livres
Jean a a.lot read of books

'Jean read a lot of books.'

Beaucoup in the nominal position and in the QAD configuration have different interpretive effects, as originally noted by Obenauer (1984). For instance, (20a) can be true either if the mayor greeted a crowd of sportsmen collectively or if many sportsmen were greeted individually. In contrast, (20b) can only be interpreted in the latter way.

[2] Doetjes (1997) glosses *de* in (19) as *of*, while Burnett (2012) leaves it unglossed.

(20) a. Le maire a salué [**beaucoup** de sportifs]
the mayor has greeted a.lot of sportsmen

‘The mayor greeted many sportsmen.’ (✓collective; ✓distributive)

b. Le maire a **beaucoup** salué de sportifs
the mayor has a.lot greeted of sportsmen

‘The mayor greeted many sportsmen.’ (✗collective; ✓distributive)

To account for this difference, Obenauer proposes the following condition on QAD:

(21) Multiplicity of Events Requirement (Obenauer 1983, 1984)
QAD requires multiple occurrences of the verbal event.

In addition to (21), however, Burnett (2012) imposes an additional constraint on QAD based on the data in (22). Unlike the adverbial example in (22a), which can involve a single book, the QAD sentence in (22b) is infelicitous if I read only two, or just a few, of my favorite books many times. Based on this data, Burnett concludes that QAD has an additional requirement stated in (23).

(22) a. J’ai **beaucoup** lu *La guerre et la paix*
I.have a.lot read DET war and DET peace

‘I read War and Peace a lot.’

b. J’ai **beaucoup** lu de livres
I.have a.lot read of books

‘I read a lot of books.’

(23) Multiplicity of Objects Requirement (Burnett 2012)
QAD sentences are only true in contexts involving many objects.

To account for both requirements in (21) and (23) on QAD, Burnett (2012) proposes a compositional analysis of QAD *beaucoup* as a binary quantifier over ⟨event, object⟩ pairs, which cannot be reduced to a unary quantifier. This contrasts with adverbial *beaucoup*, which is a unary quantifier over events. Burnett’s analysis of QAD is shown in (24).

(24) a. Let $s, t \in \mathbb{N}$ such that $0 < s, t$. For all unary relations P and binary relations R,
$beaucoup_s^{ADV}(P) = 1$ iff $|P| > s$
$beaucoup_{s,t}^{QAD}(R) = 1$ iff $beaucoup_s^{ADV}(Dom(R))$=1 $\wedge$ $beaucoup_t^{ADV}(Ran(R))$=1

b.
[VP $_t$ (c) [beaucoup$_{s,t}$^{QAD}] [VP $_{<v,et>}$ (b) [NP $_e$ je I] [VP $_{<e<v,et>>}$ (a) [VP $_{<e<e,vt>>}$ lu $\lambda x.\lambda y.\lambda e.\,read(e,y,x)$] [NP $_{<et>}$ de livres $\lambda x.book(x)$]]]]

$[\![a]\!] = \lambda y.\lambda e.\lambda x.read(e,y,x) \wedge book(x)$
(via RESTRICT; adapted from Chung and Ladusaw (2003))
$[\![b]\!] = \lambda e.\lambda x.read(e,I,x) \wedge book(x)$
$[\![c]\!] = |\{e|read(e,I,x) \wedge book(x)\}| > s \wedge |\{x|read(e,I,x) \wedge book(x)\}| > t$

Burnett assumes that French *de*-phrases in QAD denote bare properties following previous work. However, instead of introducing narrow existential quantification based on semantic incorporation (Heyd and Mathieu 2005), Burnett adopts a modified version of Restrict to compose property-denoting arguments with verbs. As Burnett explains, her version of Restrict differs from that of Chung and Ladusaw (2003) in that the restricted argument slot is shifted to the end of the argument sequence, with existential closure delayed. This is reflected in the relevant VP nodes in (24b): type $\langle e\langle e, vt\rangle\rangle$ becomes type $\langle e\langle v, et\rangle\rangle$ after restriction.

In evaluating alternative proposals, Burnett considers whether Nakanishi's (2007) analysis of Japanese floating quantifiers could be extended to French QAD to derive the Multiplicity of Objects Requirement from the Multiplicity of Events Requirement. She concludes that the homomorphism and monotonicity constraints required by Nakanishi's analysis make incorrect predictions for French QAD. As background, beyond the homomorphism introduced in the previous section, Nakanishi further proposes that measure functions in split configurations must be monotonic relative to the part-whole structure provided by the NP individuals mapped from VP events. Following Schwarzschild (2002), Nakanishi defines a measure function as monotonic relative to a domain iff it tracks the part-whole structure of the elements in that domain, as formulated in (25).[3]

(25) Nakanishi (2007)
A measure function μ is monotonic relative to a domain I iff:
(i) there are two individuals $x, y \in I$ such that x is a proper subpart of y, and
(ii) $\mu(x) < \mu(y)$.

To show that Nakanishi's analysis makes incorrect predictions for French QAD, Burnett presents examples of participant recycling. For instance, the sentence in (26) allows participant recycling: it can be true in a context where some clients are paired more than once. In other words, the number of bringing-together events can exceed the number of people involved.

(26) Pendant ma longue carrière d'entremetteuse, j'ai **beaucoup** réuni de personnes
During my long career of.matchmaker I.have a.lot brought.together de people

'During my long career as a matchmaker, I brought together many people.'

A version of the monotonicity constraint that Burnett considers is one in which the constraint applies to the verbal events. She shows that the monotonicity constraint on the events incorrectly rules out some participant recycling cases. Note, however, that Nakanishi's original account requires the constraint to apply to the individuals mapped from the events, rather than to the verbal events themselves. In addition, homomorphism alone does not rule out participant recycling. It allows many-to-one mappings, which is exactly what Nakanishi uses to explain cases where multiple atomic events are mapped to the same individual (see the example (18) in Section 2.4). In other words, Nakanishi's analysis per se does not block participant recycling. It simply does not address how to derive the Multiplicity of Objects Requirement from the Multiplicity of Events Requirement.

To summarize, Burnett's generalization is that both events and objects must have large cardinality, based on the infelicity judgment for (22b). This leads to the additional Multiplicity of Objects Requirement: QAD sentences must involve many objects. To account for this, Burnett incorporates the requirement directly into the denotation of QAD *beaucoup* by analyzing it as a binary quantifier.

3.2 Modeling quantification at a distance

This section presents my compositional analysis of *beaucoup* in QAD as an event-quantifying modifier, which shares a core semantics with its individual-quantifying counterpart but differs in the domain of quantification. Before presenting the analysis, I first show that multiple-object inferences arise pragmatically whenever supported by context. That is, the generalization in (23) is not a semantic

[3]Schwarzschild (2002) proposes that a measure function in a pseudopartitive is monotonic in the sense defined in (25), which explains why *three liters of water* is acceptable, whereas *three degrees of water* is not. μ_{volume} is monotonic relative to the NP *water*, but $\mu_{\text{temperature}}$ is not.

requirement but a pragmatic inference, as evidenced by the following data. The sentences in (27) are drawn from Doetjes (1997) and were cross-checked by an additional native speaker. Crucially, in (27b), the sentence is felicitous in a context where Jean took photos of the same two elephants many times.

(27) a. Marc a **beaucoup** écouté de disques ce weekend, mais c'était toujours le même.
Marc has a.lot listened of records this weekend but it.was always the same

'Marc listened to records a lot this weekend, but it was always the same one'

b. Jean a **beaucoup** photographié d'éléphants.
Jean has a.lot photographed of.elephants

'Jean took a lot of pictures of elephants.'

Context-dependency and cancellability are hallmarks of pragmatic inference. (27a) shows that the multiple object inference can be explicitly canceled. In (27b), the inference that Jean photographed many elephants is not obligatory; the sentence is perfectly acceptable in a context where Jean took many photos of the same two elephants. This strongly suggests that the so-called Multiple Object Requirement is not a semantic requirement and therefore should not be lexically encoded. Burnett's compositional analysis incorrectly predicts the sentences in (27) to be unacceptable in these contexts.

Under the pragmatic account, the infelicity judgment reported for *J'ai beaucoup lu de livres* 'I read a lot of books' when I read the same two books repeatedly is also expected due to its context-dependency. Indeed, similar inferences arise in other event-quantificational contexts involving numerals and measure phrases. Recall from the previous section that (28) has what has been described as an event-related reading: there may be fewer than 4,000 ships, but the total number of passages through the lock adds up to 4,000 (Krifka 1990).

(28) 4,000 ships passed through the lock last year. (✓object-related; ✓event-related)

What counts as the minimum number of ship objects in the event-related reading? The answer is that it is highly context-sensitive. Suppose I know that only two ships passed through the lock in Chicago River, each about 2,000 times. Out of the blue, it is infelicitous to utter the sentence in (28), just as Burnett reports that it is infelicitous to say *J'ai beaucoup lu de livres* when I (of course know that I) read only the same two books repeatedly. By contrast, if I am an employee in a statistics bureau that strictly tracks ship passages, the utterance of (28) becomes felicitous even in a 'two ships $\times$ 2,000 passages' scenario.

What can be concluded from all this is that the apparent infelicity of event-quantificational sentences in small-number-object contexts arises as a pragmatic inference, as supported by its context-dependent and cancellable nature. By the same reasoning, for numerals and measure phrases, if the inference is pragmatic, yielding either felicitous and infelicitous judgments depending on the context, then we do not want to analyze *4,000* as an irreducible binary quantifier that lexically encodes both a cardinality condition on events and a separate condition on the number of participants like $|x| > d$. Crucially, this binary quantifier meaning would have to exist on top of the standard unary denotation for all numerals and measure phrases.

Given this, I propose an analysis of QAD *beaucoup* that does not encode the Multiplicity of Objects Requirement in its denotation. (29) repeats the minimal pair of non-split and split quantification sentences.

(29) a. Le maire a salué [**beaucoup** de sportifs]
the mayor has greeted a.lot of sportsmen

b. Le maire a **beaucoup** salué de sportifs
the mayor has a.lot greeted of sportsmen

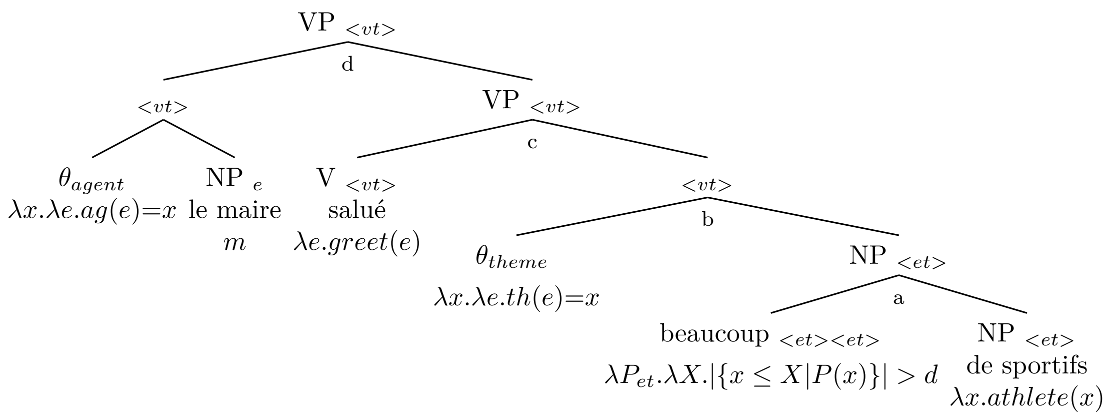
'The mayor greeted many sportsmen.'

Starting with (29a), the sentence can be true either if the mayor greeted many sportsmen collectively or if many sportsmen were greeted individually. The former scenario corresponds to a reading where many sportsmen were greeted, regardless of the number of greetings, while the latter distributive scenario is a special case of what Obenauer and subsequent work have called the 'x-times' reading, in which event multiplicity is central to the interpretation. I refer to these readings descriptively as the 'many-individuals' reading and the 'many-times' reading.

The derivations for each reading of (29a) are shown in (31) and (32): one involving $beaucoup_{ind}$ and the other involving $beaucoup_{event}$. I assume a neo-Davidsonian framework in which every argument and adjunct composes via modification, and verbs are associated with their arguments by thematic predicates. In (31), the property-denoting NP directly composes with $beaucoup_{ind}$. The resulting property then serves as the argument of the thematic predicate θ_{theme} via Chung and Ladusaw's (2003) Restrict followed by existential closure. I define $\textsc{restrict}_E$ in (30), adapted for a neo-Davidsonian framework where thematic roles are relations between individuals and events. The property argument restricts the theme predicate's individual argument X, which gets existentially bound, and the resulting event predicate continues to compose upward, accumulating further event restrictions.[4]

(30) $\textsc{restrict}_E$
If α is of type $\langle e, vt\rangle$ with denotation $\lambda x.\lambda e.R(e,x)$, and β is of type $\langle e,t\rangle$ with denotation $\lambda y.Q(y)$, then $[\![\alpha\ \beta]\!] = \lambda e.\exists x[R(e,x) \wedge Q(x)]$

(31) 'many-individuals' reading of (29a) with $beaucoup_{ind}$

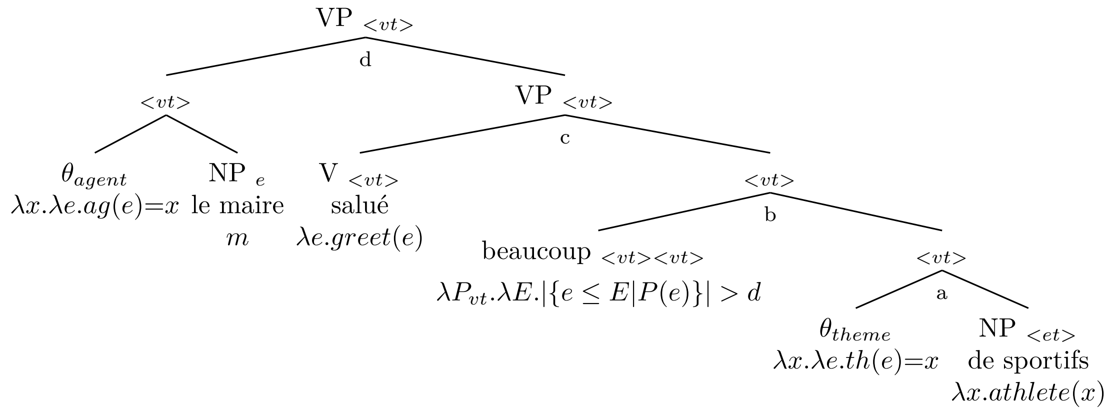


$[\![a]\!] = \lambda X.|\{x \leq X | athlete(x)\}| > d$
$[\![b]\!] = \lambda E.\exists X.th(E) = X \wedge |\{x \leq X | athlete(x)\}| > d$ (via $\textsc{Restrict}_E$)
$[\![c]\!] = \lambda E.\exists X.greet(E) \wedge th(E) = X \wedge |\{x \leq X | athlete(x)\}| > d$
$[\![d]\!] = \lambda E.\exists X.greet(E) \wedge ag(E) = m \wedge th(E) = X \wedge |\{x \leq X | athlete(x)\}| > d$
$\rightarrow \exists E.\exists X.greet(E) \wedge ag(E) = m \wedge th(E) = X \wedge |\{x \leq X | athlete(x)\}| > d$ (via $\exists_E$)

In (32), however, the NP has already has composed with the theme predicate at the time of $beaucoup_{event}$ enters the derivation. As an event modifier, $beaucoup_{event}$ targets the property of events that includes a theme description. The final truth conditions ensure that the number of subevents satisfying this description exceeds d, a contextually determined threshold for counting as many, leading to the 'many-times' reading of (29a).

[4] In Chung and Ladusaw's (2003) system, property-denoting participant arguments, which compose via RESTRICT and are therefore unsaturated, become saturated at what they call the event level, i.e., the stage of composition that reaches the event argument associated with the predicate. Adopting a Davidsonian event semantics, they note that the event argument "escapes saturation" while all other arguments are saturated at that level. Translating this into the neo-Davidsonian framework assumed here, the corresponding event level is the point at which properties combine with *thematic* predicates. The result is that property-denoting NPs restrict the individual argument of the thematic predicate and then become saturated by ∃, while the event argument remains semantically unsaturated and continues to be restricted as the composition proceeds, as defined in (30).

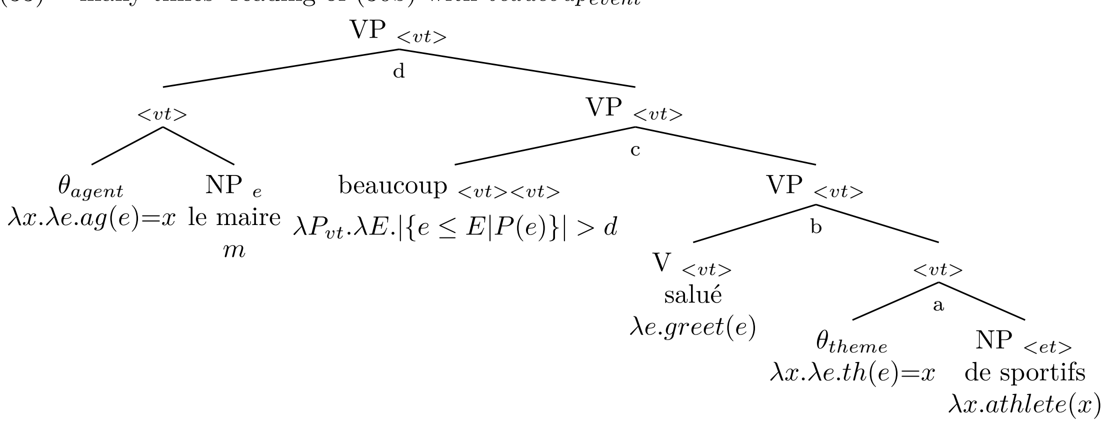
(32) ‘many-times’ reading of (29a) with *beaucoup*$_{event}$



$[\![a]\!] = \lambda e.\exists x.th(e) = x \wedge athlete(x)$ (via $\text{R}\small{\text{ESTRICT}}_E$)
$[\![b]\!] = \lambda E.|\{e \leq E|\exists x.th(e) = x \wedge athlete(x)\}| > d$
$[\![c]\!] = \lambda E.greet(E) \wedge |\{e \leq E|\exists x.th(e) = x \wedge athlete(x)\}| > d$
$[\![d]\!] = \lambda E.greet(E) \wedge ag(E) = m \wedge |\{e \leq E|\exists x.th(e) = x \wedge athlete(x)\}| > d$
$\rightarrow \exists E.greet(E) \wedge ag(E) = m \wedge |\{e \leq E|\exists x.th(e) = x \wedge athlete(x)\}| > d$ (via $\exists_E$)

Turning to split quantification, the QAD sentence (29b) only has the ‘many-times’ reading, in line with the Multiplicity of Events requirement. Consequently, it has a single compositional derivation, as shown in (33), involving the same *beaucoup*$_{event}$ as in (32), now composed at the VP level. Crucially, the semantic composition yields logically equivalent truth conditions whether beaucoup$_{event}$ is interpreted before the verb, as in (32), or after the verb, as in (33). Because QAD only occurs with distributive predicates, as will be shown in Section 3.3, for any verbal predicate V licensed in QAD, $V(E)$ is equivalent to $\bigwedge_{e \leq E} V(e)$.

(33) ‘many-times’ reading of (30b) with *beaucoup*$_{event}$



$[\![a]\!] = \lambda e.\exists x.th(e) = x \wedge athlete(x)$ (via $\text{R}\small{\text{ESTRICT}}_E$)
$[\![b]\!] = \lambda e.greet(e) \wedge \exists x.th(e) = x \wedge athlete(x)$
$[\![c]\!] = \lambda E.|\{e \leq E|greet(e) \wedge \exists x.th(e) = x \wedge athlete(x)\}| > d$
$[\![d]\!] = \lambda E.ag(E) = m \wedge |\{e \leq E|greet(e) \wedge \exists x.th(e) = x \wedge athlete(x)\}| > d$
$\rightarrow \exists E.ag(E) = m \wedge |\{e \leq E|greet(e) \wedge \exists x.th(e) = x \wedge athlete(x)\}| > d$ (via $\exists_E$)

For syntactic analyses of QAD, there are broadly two approaches: movement (e.g., Boivin 1999) and base-generation (e.g., Doetjes 1997), illustrated in (34a) and (34b).

(34) a. Movement
Jean a $[_{VP}$ beaucoup$_1$ $[_{VP}$ lu [___$_1$ $[_{NP}$ de livres]]]]

b. Base-generation
Jean a $[_{VP}$ beaucoup $[_{VP}$ lu $[_{NP}$ de livres]]]

The proposed compositional semantics is compatible with both syntactic analyses of QAD. Under the movement approach, *beaucoup*$_{event}$ in (32) moves to the pre-verbal position for an independent syntactic reason, though the exact modeling depends on one's syntactic assumptions. By contrast, *beaucoup*$_{ind}$ in (31) would result in a type mismatch in the displaced position since the VP is a predicative of events. Under the base-generation approach, *beaucoup*$_{event}$ is directly merged in the pre-verbal position and composes with the VP, as shown in (33).

In summary, *beaucoup* receives a uniform part-quantifying semantics that applies polymorphically to either the individual or event domain. As shown in (35), *beaucoup* has two denotations differing only in argument type: properties of individuals or properties of events. The core meaning based on the part-whole relation remains constant across domains. Thematic predicates play a crucial role in linking events and individuals, and their structural positions relative to the quantifier correlate with the possible readings of *beaucoup* across split and non-split configurations.

(35) a. $[\![\text{beaucoup}_{ind}]\!] = \lambda P_{et}.\lambda X.|\{x \leq X|P(x)\}| > d$

b. $[\![\text{beaucoup}_{event}]\!] = \lambda P_{vt}.\lambda E.|\{e \leq E|P(e)\}| > d$

Before proceeding, I discuss adverbial *beaucoup* that does not occur with *de*-phrases. In split quantification, *beaucoup* quantifies over discrete event parts that are individuated via the theme relation, using a cardinality function that counts atomic subevents. However, with intransitive verbs like *dormir* 'sleep' where no theme is present to provide such individuation, *beaucoup* instead involves a continuous measure function μ over event properties such as duration or intensity. Rather than counting discrete parts, this measure function compares the event's scalar property to a contextual threshold d, yielding the requirement $\mu(E) > d$ (e.g., the sleeping duration exceeds 10 hours).

(36) Jean a **beaucoup** dormi
Jean has a.lot slept

'Jean slept a lot.'

This contrast reflects a fundamental distinction between discrete quantification and continuous measurement. In discrete quantification, *beaucoup* applies cardinality-based threshold comparison over individuated atomic parts. In continuous measurement, it applies threshold comparison over continuous properties. Both share a core semantics of comparing values to a threshold, but they differ in whether the relevant domain consists of discrete, countable units or continuous, measurable properties.

One might wonder whether QAD sentences could also involve continuous measure functions, as adverbial sentences like (36) do. While this is analytically possible, adopting a measure-based approach for QAD would still require an additional mechanism to restrict the measure function to cardinality-based quantification, since QAD exclusively allows the 'many-times' reading. Indeed, this restriction is one of the defining characteristics of QAD, as suggested by Obenauer (1983). Verbs that only allow what can be described as an intensity reading with adverbial *beaucoup* 'a lot' and *peu* 'little', such as *apprécier* 'appreciate' or *inquiéter* 'worry', block QAD altogether.

(37) a. La nouvelle m'a **beaucoup** inquiété.
the news me.have a.lot worried

'The news worried me a lot.' (✓intensity; ✗many-times)

b. *La nouvelle a **beaucoup** inquiété d'experts.
the news have a.lot worried of.experts

'The news worried many experts.'

c. Le critique a **peu** apprécié le film.
the critic has little appreciated the film

‘The critic didn’t appreciate the film much.’ (✓intensity; ✗many-times)

d. *Le critique a **peu** apprécié de films.
the critic has little appreciated of films

‘The critic didn’t appreciate many films.’

Therefore, a measure-based analysis would be no more economical than distinguishing discrete quantification for QAD from continuous measurement for adverbial cases like (36), since it would still require additional mechanisms to restrict QAD to cardinality-based interpretations. While a unified measure-based semantics with appropriate restrictions might be theoretically desirable, I adopt the more straightforward cardinality-based semantics for QAD *beaucoup*.

3.3 Event-sensitive restrictions

Quantification at a distance (QAD) in French shows event-sensitive restrictions similar to those observed in Japanese floating quantifiers (Section 2.1). Specifically, collective interpretations, stative verbs, and single-event scenarios are incompatible with French QAD. I show that these restrictions follow from the proposed analysis.

3.3.1 Collective interpretations

The first restriction on split quantification of *beaucoup* is that it is not compatible with collective interpretations, as shown in (38), unless the sentence is understood as referring to many bringing-together events (in which case the interpretation is essentially distributive). The example is from Burnett (2012). Under the current analysis, (38) receives the simplified truth conditions in (39).

(38) *Pour la fête hier, j’ai beaucoup réuni de personnes
For the party yesterday, I.have a.lot gathered of people

‘I brought many people together for the party yesterday.’

(39) $\exists E.ag(E) = I \wedge |\{e \leq E | brought\text{-}together(e) \wedge \exists x.th(e) = x \wedge person(x)\}| > d$

The truth conditions in (39) count atomic subevents. However, in a collective scenario, the VP event has only one subevent, which is the collective event itself. Since the cardinality of such subevents is necessarily one, the condition $> d$ cannot be satisfied. It is implausible for the degree parameter d for many to be smaller than 1 in any context. The sentence is therefore predicted to be unacceptable, which matches the observed judgment.

3.3.2 Statives

Second, QAD *beaucoup* cannot occur with stative verbs, as illustrated in (40). The examples are from Burnett (2012), who attributes the original observation to Obenauer (1994). The intended truth conditions of (40b) are shown in (41).

(40) a. Jean a possédé [**beaucoup** de chevaux]
Jean has owned a.lot of horses

b. *Jean a **beaucoup** possédé de chevaux
Jean has a.lot owned of horses

‘Jean owned a lot of horses.’

(41) $\exists S.holder(S) = j \wedge |\{s \leq S | own(s) \wedge \exists x.th(s) = x \wedge horse(x)\}| > d$

The truth conditions in (41) require counting atomic substates. However, states are not quantized; they lack the kind of part-whole structure that would allow their parts to be discretely individuated and counted in the way that atomic events can be. The formula in (41) is thus undefined: the cardinality function cannot apply to a domain without discrete individuation. The sentence is predicted to be unacceptable.

One might wonder whether stative predicates could instead be analyzed as having stages, akin to the subevents of progressive events in the sense of Landman (1992). However, the inability to be divided and counted applies equally to the temporal parts of a state. A stage-of relation is a special kind of part-of relation, defined as follows: an event e_1 is a stage of another event e_2 if the latter can be regarded as a more developed version of the former. Intuitively, they count as the same event at different stages of development. Against this background, Landman argues that states do not have stages that count as distinct eventualities for counting purposes. The following excerpt from Landman (2008) illustrates this view:

> Now, take a state s of Fred being in Amsterdam with running time last week. And look at Fred on Monday of last week and on Friday of last week. Since Fred was in Amsterdam all of last week, there is going to be a state s_1 of Fred being in Amsterdam with running time Monday, and a state s_2 of Fred being in Amsterdam with running time Friday. That gives us three states, all with different running times, and the states s_1 and s_2 don't even temporally overlap. How many times was Fred in Amsterdam? Well, obviously only once. This means that, while the model makes s, s_1 and s_2 into three distinct eventualities, it shouldn't count them as three distinct eventualities: for the purpose of counting, they count as one (Landman 2008).

3.3.3 Single events

Lastly, *beaucoup* in the QAD configuration is infelicitous in single-event contexts. The examples in (42) are from Obenauer (1983). While *beaucoup* in its canonical position, as in (42a), is acceptable in single-event scenarios indicated by the adjunct *en soulevant le couvercle* 'lifting the lid,' *beaucoup* that appears as a split quantifier is not, as shown in (42b). (42c) creates another minimal pair with (42b), illustrating that QAD *beaucoup* becomes acceptable in many-event scenarios.

(42) a. En soulevant le couvercle, il a trouvé [**beaucoup** de pièces d'or]
in lifting the lid, he has found a.lot of coins of.gold

'Lifting the lid, he found a lot of gold coins.'

b. *En soulevant le couvercle, il a **beaucoup** trouvé [de pièces d'or]
in lifting the lid, he has a.lot found of coins of.gold

'Lifting the lid, he found a lot of gold coins.'

c. En cherchant partout, il a **beaucoup** trouvé [de pièces d'or]
in searching everywhere, he has a.lot found of coins of.gold

'Searching everywhere, he found a lot of gold coins.'

Individual-quantifying $beaucoup_{ind}$ does not require multiple subevents, since the minimal parts it counts are individuals. In contrast, $beaucoup_{event}$ involves quantification over subevents. The single lid-lifting event in (42b) is atomic, so its only subevent is itself. A cardinality of 1 is unlikely to exceed the threshold and thus fails to satisfy the truth conditions, just as in the case of collective events.

Doetjes (1997) presents the following mass-noun example as potential counterevidence to the multiple-event requirement for QAD. The objection is that this describes a single event of water-spouting rather than multiple discrete events, yet *beaucoup* remains felicitous as a split quantifier.

(43)	Pendant	ces	dix	minutes,	la	fontaine	a	**beaucoup**	craché	d'eau
	during	these	ten	minutes,	the	fountain	has	a.lot	spouted	of.water

'During these ten minutes, the fountain spouted a lot of water.'

While we might intuitively describe the scenario as a single event at a high level of granularity, what counts as an atomic event depends on the contextually relevant level of individuation. A ten-minute event of water-spouting can be viewed as a single event, but for other purposes, particularly when measuring quantity via subevents, it can decompose into multiple spouting subevents. Lid-lifting events, in principle, could similarly have parts, but the crucial difference is that the adjunct *en soulevant le couvercle* 'upon lifting the lid' enforces a coarse-grained individuation that treats the entire finding event as a single, indivisible unit, precluding access to finer-grained subevents. By contrast, *pendant ces dix minutes* 'during these ten minutes' or *en cherchant partout* 'searching everywhere' support a finer-grained individuation that makes multiple subevents accessible for counting. Thus, what counts as an atomic event is determined by a contextually salient level of granularity: sentences with QAD *beaucoup* are evaluated relative to an individuation level at which subevents are available.

In this sense, the fountain's ten-minute water-spouting event in (43) forms one plural event E at a coarse grain, but decomposes into atomic subevents e at the finer grain relevant for counting. Crucially, in my analysis, $beaucoup_{event}$ counts these atomic subevents, not the maximal plural event E. The adjunct in (42b) blocks this fine-grained individuation, making QAD infelicitous, whereas the adjuncts in (42c) and (43) support counting subevents.

3.4 Section summary

This section has examined quantification at a distance (QAD) in French, focusing on the behavior of *beaucoup* 'a lot' in split configurations. Building on Burnett's (2012) compositional account, which analyzes QAD *beaucoup* as an irreducible binary quantifier over both events and objects, I have argued that split *beaucoup* involves only event quantification. The apparent multiplicity of objects arises not from the semantics of *beaucoup* itself, but from context-dependent and cancelable inferences.

The proposed compositional analysis is grounded in a syntax-semantics interface that remains faithful to surface word order, accounting for how *beaucoup* in canonical nominal positions receives two interpretations while having only one derivation in QAD configurations. Crucially, it does so while preserving a single core semantics for *beaucoup*, with variation only in the domain of quantification: individuals and events. I have also shown that the proposed semantics is sufficient to derive the three event-sensitive restrictions on QAD: incompatibility with collective predicates, stative predicates, and single-event contexts. The analysis offers a unified account of *beaucoup* in a manner compatible with both syntactic options for QAD. In the next section, I turn to reverse proportional measure phrases, another instantiation of split quantification.

4 Reverse proportional measure phrase

4.1 Introduction

In this section, I develop a compositional account of reverse proportional measure phrases, building on event-quantificational analyses of Japanese floating quantifiers and French quantification at a distance. Under reverse proportional interpretations, the proportion appears to be calculated relative to verbal events rather than nominal objects.

I begin by reviewing previous analyses of reverse proportional quantity adjectives such as *many* and *few*. These analyses fall into three broad categories: standard-based approaches (Cohen 2001; Romero 2021), LF-mapping approaches (Herburger 1997), and ambiguity approaches (Westerståhl 1985; Bale and Schwarz 2020). Each approach offers valuable insights but also faces empirical or theoretical challenges. Reviewing both reverse proportional adjectives and measure phrases establishes the analytical landscape for situating the present approach within the broader discussion of reverse proportionality.

4.1.1 Previous analyses of reverse proportional quantity adjectives

Westerståhl (1985) Westerståhl (1985) identified what has come to be known as the reverse proportional reading of the quantity adjectives *many* and *few*. The key empirical observation was that sentences like (44) can be understood not only in the ordinary proportional sense but also in a reverse proportional manner. The latter interpretation shifts the denominator of the ratio from the restrictor of *many* to its nuclear scope.

(44) Many Scandinavians have won the Nobel prize in literature.
'Many of the Scandinavians are Nobel prize winners in literature.' (✓proportional)
'Many of the Nobel prize winners in literature were Scandinavians.' (✓reverse proportional)

Westerståhl outlines an additional reverse proportional entry for *many* besides the cardinal and ordinary proportional entries, as shown in (45c), where n and p are contextual standards of cardinality and proportion.

(45) a. $[\![many_{cardinal}]\!] = \lambda P.\lambda Q.\ |P \cap Q| > n$

b. $[\![many_{prop}]\!] = \lambda P.\lambda Q.\ \frac{|P \cap Q|}{|P|} > p$

c. $[\![many_{rev.prop}]\!] = \lambda P.\lambda Q.\ \frac{|P \cap Q|}{|Q|} > p$

The entry in (45c) captures the observed truth conditions directly, but at a theoretical cost, since it violates the Conservativity Universal.

(46) a. Conservativity universal (Barwise and Cooper 1981; Keenan and Stavi 1986):
All quantificational determiners in natural language are conservative.

b. f is conservative iff $f(P)(Q) = f(P)(P \cap Q)$.

To preserve conservativity, Westerståhl suggests a solution that relies on the context-dependent nature of the standard. In particular, the lexical entries in (45) can be rewritten as follows:

(47) a. $[\![many_{cardinal}]\!] = \lambda P.\lambda Q.\ |P \cap Q| > n$

b. $[\![many_{prop}]\!] = \lambda P.\lambda Q.\ |P \cap Q| > p \cdot |P|$

c. $[\![many_{rev.prop}]\!] = \lambda P.\lambda Q.\ |P \cap Q| > p \cdot |Q|$

Given that both n and p are contextually determined standards, the standards for proportion in (47b) and (47c), $p \cdot |P|$ and $p \cdot |Q|$, can be considered special cases where $n = p \cdot |P|$ and $n = p \cdot |Q|$, respectively. This way, *many* has a single denotation with a context-dependent standard as a threshold.

Herburger (1997) Herburger (1997) rejects the idea that reverse proportional readings require a distinct, non-conservative lexical entry. Instead, she attributes the apparent reversal in interpretation to focus association, while retaining the standard conservative proportional meaning for *many* and *few*, as shown in (48).

(48) a. $[\![many_{prop}]\!] = \lambda P.\lambda Q.\ \frac{|P \cap Q|}{|P|} > d$

b. $[\![few_{prop}]\!] = \lambda P.\lambda Q.\ \frac{|P \cap Q|}{|P|} < d$

Herburger argues that reverse proportional readings arise only when weak DPs are focused. Based on the data in (49), she concludes that the reverse proportional reading results from focus-affected quantification, with the VP in (49a) serving as the restrictor.

(49) a. Many [Scandinavians]$_F$ have won the Nobel prize in literature.
(✓reverse proportional)

b. Many Scandinavians have won [the Nobel prize in literature]$_F$.
(✗reverse proportional)

Following Rooth (1992), Herburger provides the sentence in (50a) with the LF in (50b), based on focus association and the conservative denotation in (48b).

(50) a. Few [cooks]$_F$ applied.
'Few applicants were cooks.' (✓reverse proportional)

b. [Few x [$\exists P.P(x) \wedge \text{applied}(x)$]] $\text{cooks}_F(x) \wedge \text{applied}(x)$

In the proportional reading, the syntactic restrictor is *cooks* (P), and the nuclear scope is *applied* (Q). However, when the NP *cooks* is focused, as in (50a), the effective semantic restrictor becomes the set of applicants, as shown in (50b). The resulting interpretation compares the set of cook applicants to the applicant set, yielding the reverse proportional meaning. Conservativity is maintained by locating the source of variation at the syntax-semantics interface.

Herburger points out that focus association in reverse proportional *many* and *few* parallels focus association in adverbs of quantification such as *always*, *usually*, and *rarely*. She cites the following example from Rooth (1985):

(51) a. In St. Petersburg, officers always escorted [ballerinas]$_F$.

b. In St. Petersburg, [officers]$_F$ always escorted ballerinas.

(51a) means that whenever officers escorted someone, they were ballerinas, while (51b) means that whenever ballerinas were escorted, it was by officers. As Rooth proposed, the implicit restrictor for adverbs of quantification is supplied by the sentence's focus structure, specifically with the non-focused part serving as the restrictor. Herburger's claim is that reverse proportional *many* and *few* can be analyzed in the same way that adverbs of quantification have been analyzed.

Cohen (2001) Cohen (2001) offers an account of reverse proportional readings of *many* and *few* as instances of what he calls relative readings. Cohen begins with the well-known example from Westerståhl (1985), repeated below.

(52) Many Scandinavians have won the Nobel prize in literature.
'Many winners of the Nobel prize in literature were Scandinavians.' (✓reverse proportional)

Given that 14 out of 81 Nobel laureates were Scandinavian (as of 1984, according to Westerståhl), the sentence in (52) seems intuitively true. Yet both the cardinal reading $|P \cap Q| > n$ and the proportional reading $\frac{|P \cap Q|}{|P|} > p$ predict that the sentence should be false, since the number and proportion of Scandinavian laureates are very small relative to the total population of Scandinavians. To capture the third reading, Westerståhl proposed a reverse proportional denotation $\frac{|P \cap Q|}{|Q|} > p$, where the denominator corresponds to the VP (the set of Nobel laureates). Cohen, however, argues that this non-conservative denotation cannot explain why similar sentences like (53) yield contrasting truth values.

(53) Many Andorrans have won the Nobel Prize in Literature.

Imagine a context in which three laureates out of roughly 100 were Andorrans and three were Scandinavians. In that case, $\frac{|P \cap Q|}{|Q|}$ is the same (three laureates out of 100), yet (53) is judged true while (52) is judged false. The same problem arises for Herburger (1997), whose conservative denotation combined with focus association fails to capture this contrast.

Based on this observation, Cohen distinguishes two interpretations available for Westerståhl's example as *absolute* and *relative* proportional readings. In the absolute proportional reading, the proportion of Scandinavian Nobel laureates among Scandinavians exceeds some contextually determined standard p. In the relative proportional reading, by contrast, p corresponds to the proportion of Nobel laureates in literature within a broader comparison class (e.g., the world population). Crucially, in the relative reading, a set of alternatives $ALT(P)$ for the property P plays a key role. For instance, $ALT(\mathit{Scandinavian}) = \{\mathit{Scandinavian}, \mathit{English}, \mathit{French}, \ldots\}$. Cohen defines a set of alternatives A that combines alternatives to both the restrictor and the nuclear scope: $A = \{P' \cap Q' \mid P' \in ALT(P) \text{ and } Q' \in ALT(Q)\}$. The denotation of proportional *many* is as follows:

(54) $\mathit{many}(P)(Q)$ is true iff $\frac{|P \cap Q|}{|P \cap \bigcup A|} > p$, where

a. p is large (absolute reading), or

b. $p = \frac{|\bigcup A \cap Q|}{|\bigcup A|}$ (relative reading).

Cohen shows that the relative reading gives rise to apparent reversal because, under this interpretation, *many* becomes symmetric. To see this symmetry, note that under the relative reading, $\mathit{many}(P)(Q)$ is true iff:

$$\frac{|P \cap Q|}{|P \cap \bigcup A|} > \frac{|\bigcup A \cap Q|}{|\bigcup A|}.$$

This is equivalent to

$$\frac{|P \cap Q|}{|Q \cap \bigcup A|} > \frac{|\bigcup A \cap P|}{|\bigcup A|},$$

which is exactly the truth condition for $\mathit{many}(Q)(P)$ with a relative contextual parameter.

Applying this to the LF *many(Scandinavians)(have won the Nobel Prize in literature)*, the truth conditions come out as:

$$\frac{|\mathit{Scandinavian} \cap \mathit{Nobel}|}{|\mathit{Scandinavian}|} > \frac{|\mathit{Nobel}|}{|\mathit{population}|},$$

which says that the proportion of Nobel laureates among Scandinavians is greater than the proportion of Nobel laureates among the world population. By the symmetry shown above, this is equivalent to the truth conditions of the LF *many(have won the Nobel Prize in literature)(Scandinavians)*:

$$\frac{|\mathit{Scandinavian} \cap \mathit{Nobel}|}{|\mathit{Nobel}|} > \frac{|\mathit{Scandinavian}|}{|\mathit{population}|}.$$

Crucially, the relative formulation derives a symmetry such that $\mathit{many}(P, Q) \iff \mathit{many}(Q, P)$ rather than stipulating it. Proportional and reverse proportional readings are therefore two instances of a single schema. The apparent argument reversal follows from how the contextual standard is computed. Specifically, when the NP bears contrastive topic marking, it activates a set of relevant alternatives, licensing the relative comparison.

Romero (2021) Romero (2021) builds on Cohen's (2001) characterization of the reverse proportional reading and yet argues that it is still empirically inadequate. The problem is that Cohen's truth conditions compare the proportion $|P \cap Q| : |P|$ only to the aggregate proportion $|\bigcup \mathrm{ALT}(P) \cap Q| : |\bigcup \mathrm{ALT}(P)|$, but fail to account for the distribution of proportions across alternatives. For instance, consider the following sentence.

(55) Many Dutch_{ALT} scholars got an ERC grant.

In scenario 1, the proportions for most countries cluster between 5‰ and 6‰, making Holland's 7‰ stand out as relatively high. Accordingly, the sentence is judged true. In scenario 2, by contrast, the proportions range between 5‰ and 8‰ and peak at 8‰, making Holland's 7‰ not impressive, and the sentence is judged false. In both scenarios, the relevant proportions remain constant: 7 out of 1,000 Dutch scholars received grants (7‰), and overall, 156 out of 30,000 scholars in ERC territory received grants (5.2‰). However, the two scenarios differ in how the proportions are distributed across individual countries. Since Cohen's formulation references only the aggregate proportion and not the point-wise distribution of alternative proportions, it incorrectly predicts the same truth value for both scenarios.

To address this, Romero revises the truth conditions by incorporating the set of alternative proportions $\{|P' \cap Q| : |P'| \mid P' \in \text{ALT}(P)\}$ and deriving a threshold value based on their distribution. The revised truth condition for the reverse proportional reading is given in (56):

(56) $|P \cap Q| : |P| > \theta(\{|P' \cap Q| : |P'| \mid P' \in \text{ALT}(P)\})$

Here, the function θ takes the set of alternative proportions and computes a threshold value (e.g., via mean or other statistical measures). This formulation captures the intuition that whether a given proportion counts as many depends on how it compares to the distribution of proportions across point-wise alternatives.

Romero claims that the revised truth conditions in (56) require *many* to be non-conservative since $|P' \cap Q| : |P'|$ for each $P' \in \text{ALT}(P)$ would involve non-Scandinavians. To maintain conservativity, Romero decomposes *many* into two components, building on Hackl (2000) and Solt (2009): a conservative, parametrized determiner stem $\text{MANY}_{\text{prop}}$ and the positive degree operator POS. Under her analysis, it is POS that is non-conservative. The denotations are given in (57):

(57) a. $[\![\text{MANY}_{\text{prop}}]\!] = \lambda d.\lambda P.\lambda Q.(|P \cap Q| : |P|) \geq d$

b. $[\![\text{POS}]\!] = \lambda C_{\langle dt,t\rangle}.\lambda D_{\langle dt\rangle}.L(C) \subseteq D$

The stem $\text{MANY}_{\text{prop}}$ is conservative and takes a degree argument d. The operator POS combines with a comparison class C (a set of degree sets) and returns a predicate that holds of a degree set D just in case D is a superset of $L(C)$ derived from the comparison class. Applied to a comparison class, L returns a set of degrees representing the neutral interval for that class. This interval provides a standard against which other degree sets can be compared, letting POS to determine whether a given degree set exceeds typical values.

The comparison class C, and thus whether the reading is reverse or non-reverse, depends on which element in the sentence serves as the source of alternatives. The operator POS is associated with some element X_{ALT} in the sentence, and the comparison class is constrained to correspond to the meaning of POS's syntactic sister relative to X_{ALT}. When X_{ALT} is external to the host NP, the regular proportional reading arises. When X_{ALT} is internal to the host NP, the reverse proportional reading occurs. For instance, in the proportional reading of (58), POS associates with an element external to the host NP. The LF and derivations are shown in (59).

(58) Many faculty children had a good$_{\text{ALT}}$ time.

(59) a. LF: [[POS C] [1 [[t_1-$\text{MANY}_{\text{prop}}$ faculty children] had a good$_{\text{ALT}}$ time]] $\sim C$]

b. $[\![$1 [t_1-$\text{MANY}_{\text{prop}}$ faculty children had a good time] $]\!] =$
$\lambda d.(|\{x : \text{f-child}(x)\} \cap \{x : \text{have-good-time}(x)\}| : |\{x : \text{f-child}(x)\}|) \geq d$

c. $[\![C]\!] \subseteq \{\lambda d'.(|\{x : \text{f-child}(x)\} \cap \{x : \text{have-good-time}(x)\}| : |\{x : \text{f-child}(x)\}|) \geq d',$
$\lambda d'.(|\{x : \text{f-child}(x)\} \cap \{x : \text{have-bad-time}(x)\}| : |\{x : \text{f-child}(x)\}|) \geq d',$
$\lambda d'.(|\{x : \text{f-child}(x)\} \cap \{x : \text{have-regular-time}(x)\}| : |\{x : \text{f-child}(x)\}|) \geq d',$
$\ldots\}$

d. $[\![(59\text{a})]\!] = L([\![C]\!]) \subseteq \lambda d.(|\{x : \text{f-child}(x)\} \cap \{x : \text{have-good-time(x)}\}| : |\{x : \text{f-child}(x)\}|) \geq d$

In (59a), POS scopes out of its host NP and associates with *good*$_{\text{ALT}}$ in the VP. The comparison class C contains alternative degree sets varying the quality of time the faculty children had. The resulting truth conditions state that the proportion of faculty children who had a good time exceeds the neutral segment derived from this comparison class.

Romero's analysis of the reverse proportional reading is provided below.

(60) Many Scandinavians$_{\text{ALT}}$ have won the Nobel Prize in literature.

(61) a. LF: [[POS C] [1 [[t_1-MANY$_{\text{prop}}$ Scandinavians$_{\text{ALT}}$] have won NP]] $\sim C$]

b. $[\![$1 [t_1-MANY$_{\text{prop}}$ Scandinavians have won NP] $]\!] =$
$\lambda d.(|\{x : \text{Scandinavian}(x)\} \cap \{x : \text{NP-winner}(x)\}| : |\{x : \text{Scandinavian}(x)\}|) \geq d$

c. $[\![C]\!] \subseteq \{\lambda d'.(|\{x : \text{Scandin}(x)\} \cap \{x : \text{NP-winner}(x)\}| : |\{x : \text{Scandin}(x)\}|) \geq d',$
$\lambda d'.(|\{x : \text{Mediterr}(x)\} \cap \{x : \text{NP-winner}(x)\}| : |\{x : \text{Mediterr}(x)\}|) \geq d',$
$\lambda d'.(|\{x : \text{MEastern}(x)\} \cap \{x : \text{NP-winner}(x)\}| : |\{x : \text{MEastern}(x)\}|) \geq d',$
$\ldots\}$

d. $[\![(61a)]\!] = L([\![C]\!]) \subseteq \lambda d.(|\{x : \text{Scandin}(x)\} \cap \{x : \text{NP-winner}(x)\}| : |\{x : \text{Scandin}(x)\}|) \geq d$

In this case, POS associates with *Scandinavians*$_{\text{ALT}}$, which is internal to the host NP. The comparison class C now contains degree sets corresponding to the proportions of Nobel Prize winners among different nationalities. The truth conditions in (61d) are equivalent to the θ-based one, shown in (62). They both state that the proportion of Scandinavians who are Nobel winners exceeds a threshold derived from the distribution of such proportions across alternative nationalities. In particular, the function L, which, when applied to the comparison class C, returns the so-called neutral segment for that C, corresponds to the function θ in (62), with both applying to the same set of pointwise alternative proportions.

(62) $|\{x : \text{Scandinavian}(x)\} \cap \{x : \text{NP-winner}(x)\}| : |\{x : \text{Scandinavian}(x)\}|$
$> \theta(\{|P' \cap \{x : \text{NP-winner}(x)\}| : |P'| \mid P' \in \text{ALT}(\{x : \text{Scandinavian}(x)\})\})$

The key contribution of Romero (2021) is that the context-dependence of the threshold is formally derived from independently motivated mechanisms for degree constructions. Whether a reading is reverse or not depends on whether X_{ALT} is internal or external to the host NP. The determiner stem MANY$_{\text{prop}}$ remains conservative throughout, while the apparent non-conservativity is shifted to POS, which is context-dependent by nature and independently needed for degree expressions.

Bale and Schwarz (2020) Bale and Schwarz (2020) revisit reverse proportionality in degree constructions where reference to contextually determined standards is eliminated, and challenge the view that reverse proportional readings arise from a context-dependent standard. They begin by distinguishing the forward proportional reading ($|P \cap Q| : |P| > p$) from the reverse proportional reading ($|P \cap Q| : |Q| > p$), noting that the latter occurs not only in positive constructions but also in comparatives, equatives, and degree questions where no contextual standard is available. Consider the comparative in (63):

(63) More cooks applied to our program than to yours.

This sentence can be true when the proportion of cooks among our applicants is greater than that among yours, even if the total number of cooks or applicants differs. The interpretation involves comparing two fixed ratios given in the discourse, suggesting that the reverse proportional meaning is available without a contextually supplied standard of comparison. According to the standard-based approach, reverse proportional readings arise from manipulations of the contextual standard. We saw its simplest variant, as originally suggested by Westerståhl (1985), $|P \cap Q| > n$, where n can be $p \times |P|$ or $p \times |Q|$, as well as Romero's (2021) decomposition account that relies on an inherent context-dependence of the POS operator.

Bale and Schwarz argue that such standard-based accounts undergenerate: in comparatives like (63), the semantics of the comparative operator *-er* removes reference to any contextual standard. Yet the reverse proportional reading remains, suggesting that another mechanism must produce it. Building on Wellwood (2015), their central proposal is that the lexical semantics of *many* and *few* contains an underspecified measure function μ, whose domain and scale vary depending on the context. Their denotation of *many* is given in (64), where $\sqcup$ is a function that maps sets of elements to their supremum, i.e., the element that represents the join of all members of the set.

(64) $[\![\text{MANY}]\!] = \lambda d.\lambda P.\lambda Q.\mu(\sqcup(P \cap Q)) \geq d$

The measure function μ can range over the settings in (65). Cardinal readings and proportional readings arise from the settings (65a) and (65b), respectively, for both the *than*-clause and main clause. Reverse proportional readings arise when the measure function is instantiated as (65c). The flexibility of μ thus derives the full range of interpretations without positing multiple lexical entries for *many* or relying on contextual threshold adjustment.

(65) a. $\mu_{card} = \lambda x.|x|$

b. $\mu_{prop} = \lambda x.|x| : |\sqcup [\![\text{cooks}]\!]|$

c. $\mu_{rev.prop1} = \lambda x.|x| : |\sqcup \{x : x \text{ applied to your program}\}|$ (*than*-clause)
$\mu_{rev.prop2} = \lambda x.|x| : |\sqcup \{x : x \text{ applied to our program}\}|$ (main clause)

Their LF of the comparative sentence (63) is provided in (66a). Interpreting *-er* as in the standard degree analysis (66b) and instantiating μ as $\mu_{rev.prop}$, we obtain the truth conditions in (66c), i.e., the ratio of cooks to total applicants in our program exceeds that in yours. No context-dependent standard is invoked.

(66) a. -er λd[than [[[d MANY] cooks] λx[x applied to your program]]]
λd [[[d MANY] cooks] λx[x applied to our program]]]

b. $[\![\text{-}er]\!] = \lambda D_{2\langle dt\rangle}.\lambda D_{1\langle dt\rangle}.\max(D_1) > \max(D_2)$

c. $[\![(66\text{a})]\!] = \frac{|\sqcup\{x:x \text{ cooks}(x)\}\cap\{x:x \text{ applied to our program}\}|}{|\sqcup\{x:x \text{ applied to our program}\}|} > \frac{|\sqcup\{x:x \text{ cooks}(x)\}\cap\{x:x \text{ applied to your program}\}|}{|\sqcup\{x:x \text{ applied to your program}\}|}$

Although both Bale and Schwarz (2020) and Romero (2021) adopt degree-based approaches, their explanatory centers differ. In Romero's account, the crucial mechanism is the comparison-class argument of POS, whose internal versus external anchoring gives rise to the reversal. As Bale and Schwarz note, however, her account faces difficulties in explaining reverse proportional readings in standard-fixing constructions. In contrast, for Bale and Schwarz, the reversal originates in the measure function rather than in the standard: the measure function may refer to either the restrictor or the scope set when computing the denominator. This allows their analysis to account for reverse proportional readings in a wider range of degree constructions such as comparatives, where contextual standards are fixed. Nevertheless, as Bale and Schwarz themselves acknowledge, their proposal closely resembles a lexical ambiguity approach, since the variation is attributed to the semantics of MANY itself, essentially resulting in a non-conservative treatment.

4.1.2 Previous analyses of reverse proportional measure phrases

Reviewing the various analytical options proposed for reverse proportional adjectives, three major approaches emerge: the standard-based analysis (Cohen 2001; Romero 2021), the LF-mapping analysis (Herburger 1997), and the ambiguity analysis (Westerståhl 1985; Bale and Schwarz 2020). Across these approaches, the central challenge has been to account for the full empirical range of reverse proportional readings while maintaining a theoretical constraint (i.e., the conservativity hypothesis) and a principled mapping between structure and meaning. Each analysis offers distinct insights but also faces weaknesses in meeting one or more of these challenges, resulting in a trade-off of strengths and limitations.

Building on this background, another instance of reverse proportionality beyond *many* and *few* arises with measure phrases such as *50%*, as observed by Ahn and Sauerland (2017), shown in (67).

(67) a. The company hired **50%** of the women.
'50% of the women were hired by the company.' (✓proportional)
'50% of the hires were women.' (✗reverse proportional)

b. The company hired **50%** women.
'50% of the women were hired by the company.' (✗proportional)
'50% of the hires were women.' (✓reverse proportional)

Standard-based accounts are not well suited for standard-fixing constructions of this kind. Between the mapping and ambiguity approaches, most existing analyses derive the reverse proportional interpretation of *50%* from a particular LF configuration triggered by focus association (Ahn and Sauerland 2017; Li 2022; Pasternak and Sauerland 2022). The available reading of (67b) is called reverse proportional because the first argument that *50%* combines with is not its local constituent *women* but instead comes from the verb *hire*. Ahn and Sauerland (2017) characterize this as a non-conservative interpretation, since in this case the measure phrase *50%* does not satisfy conservativity. In what follows, I use the more neutral term reverse proportional to refer to the reading available in (67b).

To account for sentences like (67b), Ahn and Sauerland (2017) propose that reverse proportional readings arise when the NP following the measure phrase (*women*) bears focus. The measure phrase *50%* undergoes focus movement at LF, and its restrictor is supplied contextually, bound to the focus alternatives of *women*. This gives rise to the illusion that 50% quantifies over its second argument rather than its first. To make such structures interpretable, they assume a modified trace conversion rule. Abstracting away from technical details, their proposal can be schematized as in (68).

(68) [50% C] [∼C [λx the company hired [x women$_{\mathbf{F}}$]]]

The focus-based compositional account has been proposed for other languages, such as German (Pasternak and Sauerland 2022) and Mandarin (Li 2022). Although the analyses differ in their details, they share the core claim that reverse proportional readings of *50%* arise from focus movement, which compositionally yields the reverse proportional interpretation.

In support of the displacement derivation in (68), Ahn and Sauerland (2017) show that similar reverse proportional readings arise in Korean, as in (69b), where the measure phrase appears clearly outside the corresponding NP. Such examples are taken to reflect the underlying scope-taking of the measure phrase illustrated in (68), made overt in (69b).

(69) a.

hoysa-ka	[yeca-uy	**50%**]-lul	ceyyonghayssta
company-NOM	women-GEN	50%-ACC	hired

‘The company hired 50% of the women.’ (✓proportional)

b.

hoysa-ka	yeca-lul	**50%**	ceyyonghayssta
company-NOM	women-ACC	50%	hired

‘50% of the hires were women.’ (✓reverse proportional)

Although focus-based analyses successfully derive the truth conditions of examples like (69b) based on the assumed LFs, there are a few reasons why focus-based derivations may not be on the right track to account for reverse proportional measure phrases. First, focus-based accounts require that the associated NP bears focus in order for reverse proportional interpretations to arise. For instance, in (70), *haksayng* ‘student’ must be focused.

(70) Focus analysis of reverse proportional measure phrases

[haksayng$_F$ ___$_1$]	-i	**50%**$_1$	chamsekhayssta
student	-NOM	50%	attended

‘50% of the attendees were students.’ (✓reverse proportional)

Korean marks focus more subtly through phrase boundaries and prosodic compression, which makes focus effects harder to detect than in stress-accent languages. However, focus intervention effects (Beck and Kim 1997) offer a useful diagnostic for focus association. Since focus-based analyses argue that the NP must receive focus for reverse proportional readings to arise, they predict that focus intervention effects should occur in wh-questions involving a reverse proportional measure phrase. This prediction is not borne out. With (71a) as a baseline, (71b) can still be interpreted reverse proportionally, despite the putative intervention configuration.

(71) a. *haksayng$_F$-man enu phathi-lul chamsekhayss-ni?
student-only which party-ACC attended-ITR
'Which party did only students attend?'

b. haksayng$_F$-i {enu phathi-lul} {50%} chamsekhayss-ni?
student-NOM which party-ACC 50% attended-ITR
'Which party had 50% of its attendees as students?' (✓reverse proportional)

By contrast, in English, focus intervention effects are harder to test due to wh-movement. Nevertheless, the focused element receives prosodic prominence through pitch accent and increased duration, making focal stress relatively straightforward to identify and test. Against this background, the reverse proportional reading readily arises in (72), regardless of which constituent bears prosodic prominence, whether subject, verb, or object NP, just as its partitive counterpart exhibits a proportional reading regardless of focus.

(72) The company hired 50% women. (✓reverse proportional)

Moreover, reverse proportional measure phrases cannot occur with stative verbs and adjectives, and it is not clear how focus association and movement derivations could be sensitive to predicate type. (73b) illustrates that canonical stative adjectives are incompatible with a reverse proportional interpretation of *50%*, unless they are coerced into eventive progressives, as in 'are acting aggressively.'

(73) a. [hama-uy **50%**]-ka kongkyekcekita
hippo-GEN 50%-NOM aggressive

'50% of the hippos are aggressive.' (✓proportional)

b. hama-ka **50%** kongkyekcekita
hippo-NOM 50% aggressive

'50% of the aggresive ones are hippos.' (✗reverse proportional)

Building on Ahn and Sauerland's (2017) empirical observations, I propose an alternative account of reverse proportional measure phrases, in which measure phrases quantify over subevents while varying systematically in which set serves as the reference point. The analysis to be proposed can be viewed as a variant of the ambiguity approach, where the major objection concerns a potential violation of the theoretical constraint known as the Conservativity Universal. However, this line of analysis regains appeal in light of recent work on weak conservativity by Zuber and Keenan (2019), who argue that a principled relaxation of the classical constraint remains restrictive enough to rule out almost all unattested determiner meanings while still permitting non-conservative denotations.

4.2 Modeling reverse proportional measurement

The analysis of reverse proportional measurement in split quantification is built on two main components, as in the previous section for French *beaucoup*. First, split quantifiers are polymorphic: they denote the same part-whole-based function, applying uniformly to properties of individuals and properties of events. Second, the structural position of NP arguments relative to thematic predicates correlates with each quantification: when the NP combines with the quantifier before the thematic predicate applies, individual quantification results; when the split quantifier applies after the NP combines with the thematic predicate, event quantification emerges. In Korean, reverse proportional measure phrases can associate with agents, themes, goals, as well as temporal and locative adjuncts. A compositional semantics is provided for each of these constructions below.

4.2.1 Agent

(74) illustrates proportional measurement sentences in Korean, where the measure phrase *50%* is associated with the agent-denoting subject. For descriptive purposes, I refer to minimal pairs of this kind as subject-oriented measurement sentences, given the explicit subject association in (74a). Specifically, I label (74a) as subject-oriented proportional and (74b) as subject-oriented reverse proportional.

(74a) uses genitive case as a partitive marker and receives a proportional reading. In contrast, the measure phrase in (74b) is separated from the subject by nominative case and is interpreted reverse proportionally.[5]

(74) a.

[hoywen-uy	**50%**]-ka	censi-lul	kwanlamhayssta
member-GEN	50%-NOM	exhibition-ACC	viewed

‘50% of the members viewed the exhibition.’ (✓proportional)

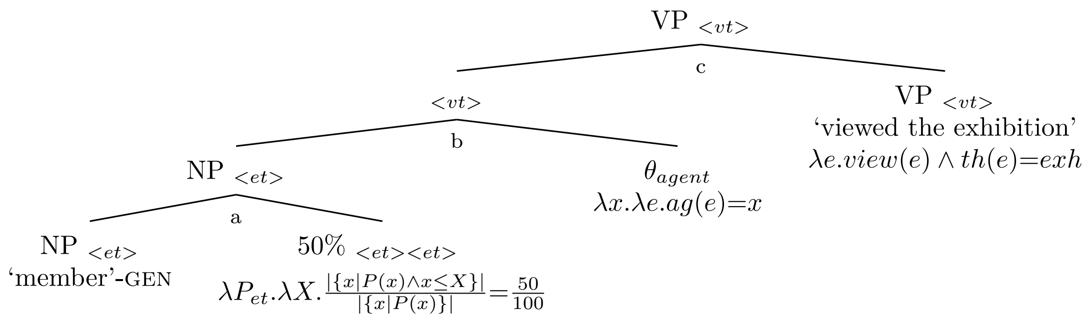
b.

hoywen-i	**50%**	censi-lul	kwanlamhayssta
member-NOM	50%	exhibition-ACC	viewed

‘50% of the exhibition’s viewers were members.’ (✓reverse proportional)

I propose that the measure phrase in (74a) is a modifier taking properties of individuals, whereas the one in (74b) takes properties of events. (75) shows the structure and composition of (74a). As in the previous section, I assume a neo-Davidsonian framework in which every argument and adjunct composes via modification, and verbs are associated with their arguments by thematic predicates. The genitive-marked NP, which denotes a property of individuals, composes with *50%.* The resulting property then serves as the argument of the thematic predicate θ_{agent}.

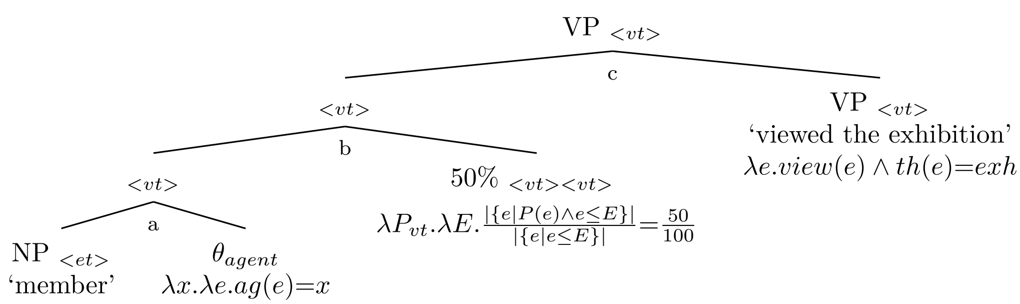
(75) Proportional measure phrase



$$[\![a]\!] = \lambda X.\frac{|\{x|member(x) \wedge x \leq X\}|}{|\{x|member(x)\}|} = \frac{50}{100}$$
$$[\![b]\!] = \lambda E.\exists X.ag(E) = X \wedge (\frac{|\{x|member(x) \wedge x \leq X\}|}{|\{x|member(x)\}|} = \frac{50}{100}) \text{ (via R}\text{ESTRICT}_E\text{)}$$
$$[\![c]\!] = \lambda E.\exists X.ag(E) = X \wedge view(E) \wedge th(E) = exh \wedge (\frac{|\{x|member(x) \wedge x \leq X\}|}{|\{x|member(x)\}|} = \frac{50}{100})$$
$$\rightarrow \exists E.\exists X.ag(E) = X \wedge view(E) \wedge th(E) = exh \wedge (\frac{|\{x|member(x) \wedge x \leq X\}|}{|\{x|member(x)\}|} = \frac{50}{100}) \text{ (via } \exists_E\text{)}$$

By contrast, the measure phrase in (74b) applies to properties of events, as shown in (76). The NP composes directly with the agent thematic predicate via Restrict$_E$ defined in Section 3. The subject-oriented reverse *50%,* as an event modifier, targets properties of events to which agent descriptions have been added.

(76) Reverse proportional measure phrase



[5] I set aside here the fact that (74b) also allows a proportional reading (‘50% of the members viewed the exhibition’), which I return to in Section 4.4.

$[\![a]\!] = \lambda e.\exists x.ag(e) = x \wedge member(x)$ (via $\textsc{Restrict}_E$)
$[\![b]\!] = \lambda E.\frac{|\{e|\exists x.ag(e)=x\wedge member(x)\wedge e\leq E\}|}{|\{e|e\leq E\}|} = \frac{50}{100}$
$[\![c]\!] = \lambda E.view(E) \wedge th(E) = exh \wedge \frac{|\{e|\exists x.ag(e)=x\wedge member(x)\wedge e\leq E\}|}{|\{e|e\leq E\}|} = \frac{50}{100}$
$\rightarrow \exists E.view(E) \wedge th(E) = exh \wedge \frac{|\{e|\exists x.ag(e)=x\wedge member(x)\wedge e\leq E\}|}{|\{e|e\leq E\}|} = \frac{50}{100}$ (via $\exists_E$)

Beyond the domain of quantification, the main difference between $\%_{ind}$ in (75) and $\%_{event}$ in (76) is that the former calculates the proportion relative to the local argument P, whereas the latter does so relative to the non-local (second) argument. In principle, proportional relations can be construed in two ways: as the proportion of A that is also B ($\frac{A\wedge B}{A}$), or as the proportion of B that is also A ($\frac{A\wedge B}{B}$). Both construals are well-defined. This is comparable to what Bale and Schwarz (2020) describe as the underspecification of measure functions in their analysis of reverse proportional adjectives. Instead, I refer to these construals as conservative and non-conservative, respectively, to distinguish these logical possibilities. These terms will also be useful for differentiating this denotational contrast from the interpretive distinction I have been using, namely proportional and reverse proportional.

Given the two quantification domains and the two construals, there are four logically possible denotations of %.

(77) a. $[\![\%_{ind.con}]\!]= \lambda d.\lambda P.\lambda X.\frac{|\{x|P(x)\wedge x\leq X\}|}{|\{x|P(x)\}|} = \frac{d}{100}$ (in (75))

b. $[\![\%_{ind.noncon}]\!]= \lambda d.\lambda P.\lambda X.\frac{|\{x|P(x)\wedge x\leq X\}|}{|\{x|x\leq X\}|} = \frac{d}{100}$

c. $[\![\%_{event.con}]\!]= \lambda d.\lambda P.\lambda E.\frac{|\{e|P(e)\wedge e\leq E\}|}{|\{e|P(e)\}|} = \frac{d}{100}$

d. $[\![\%_{event.noncon}]\!]= \lambda d.\lambda P.\lambda E.\frac{|\{e|P(e)\wedge e\leq E\}|}{|\{e|e\leq E\}|} = \frac{d}{100}$ (in (76))

Compare $\%_{ind.con}$ (77a) and $\%_{event.con}$ (77c), and similarly $\%_{ind.noncon}$ (77b) and $\%_{event.noncon}$ (77d). Although the domains to which the cardinality function applies differ, namely individuals and events, the two meanings in each pair are systematically related, resulting in parallel denotations of %. This constitutes an instance of polymorphism, characterized by a single denotation that can be instantiated at different types. By contrast, (77a) and (77b) exemplify ambiguity, depending on whether the denotation is conservative or non-conservative. The same holds for the pair (77c) and (77d).

The possibility of different construals in each domain raises the question of whether % can bear the meanings illustrated in (77b) and (77c). In Section 4.5, I propose that % indeed has the meaning in (77b) when used attributively. As for the construal in (77c), it can be used as the meaning of reverse % in a structure where the measure phrase 50% is merged in the verbal projection, with the verbal event serving as the local argument P. This is an analytical option I previously pursued in Kim (2024), though in a Davidsonian framework. However, that analysis, originally developed for Korean, does not extend well to object-oriented reverse proportional sentences in English, which will be discussed in Section 4.2.2. It would require us to adopt a structure such as *Nari* [[*bought 50%*] [*organic products*]]. But coordination-based constituency tests favor *Nari bought* [*50% organic products*]. Additionally, the analysis there treated object-oriented reverse proportional 50% on a par with secondary predicates, but in English, secondary predicates almost always follow the object. Therefore, to assign the same meaning to subject-oriented and object-oriented reverse 50% across Korean and English, I assume that both receive the denotation in (77d) with the structure presented in (76).[6]

Before proceeding to object-oriented measure phrases, I introduce an additional assumption necessary to derive the correct truth conditions for the proportional and reverse-proportional sentences discussed so far. The sentences and their final truth conditions are repeated below for convenience.

[6]As observed by Ahn and Sauerland (2017), English, unlike other languages, allows only object-oriented reverse proportional sentences, not subject-oriented ones. I leave the language-specific restrictions on subject orientation for future research.

(78) a. [hoywen-uy **50%**]-ka censi-lul kwanlamhayssta
member-GEN 50%-NOM exhibition-ACC viewed

'50% of the members viewed the exhibition.' (✓proportional)

b. hoywen-i **50%** censi-lul kwanlamhayssta
member-NOM 50% exhibition-ACC viewed

'50% of the exhibition's viewers were members.' (✓reverse proportional)

(79) a. $[\![(78a)]\!] = \exists E.\exists X.ag(E) = X \wedge view(E) \wedge th(E) = exh \wedge (\frac{|\{x|member(x)\wedge x\leq X\}|}{|\{x|member(x)\}|} = \frac{50}{100})$

b. $[\![(78b)]\!] = \exists E.view(E) \wedge th(E) = exh \wedge \frac{|\{e|\exists x.ag(e)=x\wedge member(x)\wedge e\leq E\}|}{|\{e|e\leq E\}|} = \frac{50}{100}$

In both cases, the existentially bound event variable E must denote the maximal event comprising all relevant exhibition-viewing events. Without this assumption, the default existential closure in (79a) would incorrectly allow smaller plural events to satisfy the conditions. For example, suppose 70% of the members viewed the exhibition. A smaller plural event E consisting of only 50% of these viewings could make (79a) true, despite the sentence being false in that context. The same problem arises for (79b): if there were 100 exhibition-viewing events total (50 by members, 50 by non-members), existential closure could pick out a smaller event containing only the 50 non-member viewings, incorrectly predicting the sentence with 0% as true.

The maximality issue arises generally in event-quantificational semantics with numerals and measure phrases, not only in proportional constructions. This has independently motivated event max operators for quantificational cases (Krifka 1989; Landman 2000). In what follows, I also assume reference to maximal events for measurement sentences. For simplicity, one way to enforce event maximality is to replace default event closure with a max operator for events. Building on Link (1983), the max operator can be defined as in (80):

(80) $\sigma(P) = \iota e[{}^*P(e) \wedge \forall e'({}^*P(e') \rightarrow e' \leq e)]$

This operator applies straightforwardly to sentences like (78a). For (78b), the event property that involves the measure phrase has to take scope at LF and composes with the maximal event returned by σ, as illustrated in (81). This is because, in non-conservative denotations like (77b) and (77d), both the numerator and denominator vary as functions of the same variable (X in the individual domain, E in the event domain). By contrast, in conservative denotations like (77a) and (77c), the denominator is only determined by the property argument P (e.g., $\{x|member(x)\}$) and is independent of which X or E satisfies the formula.

(81) a. [50% members] σ [$_{\text{VP2}}$ [___] [$_{\text{VP1}}$ viewed exhibition]]

b. $[\![(81a)]\!] = \frac{|\{e|e\leq E^*\wedge\exists x.ag(e)=x\wedge member(x)\}|}{|\{e|e\leq E^*\}|} = \frac{50}{100}$ (where $E* = \sigma([\![\text{VP}_2]\!])$)

4.2.2 Theme

Proportional measure phrases can also be associated with theme arguments, most commonly realized as objects in transitive sentences. (82) illustrates cases where *50%* is associated with accusative-marked transitive objects. In split quantification configurations, such as (82b), the object and the measure phrase are separated by the accusative case marker instead of the nominative.

(82) a. Nari-ka [yukinong ceyphum-uy **50%**]-lul sassta
Nari-NOM organic product-GEN 50%-ACC bought

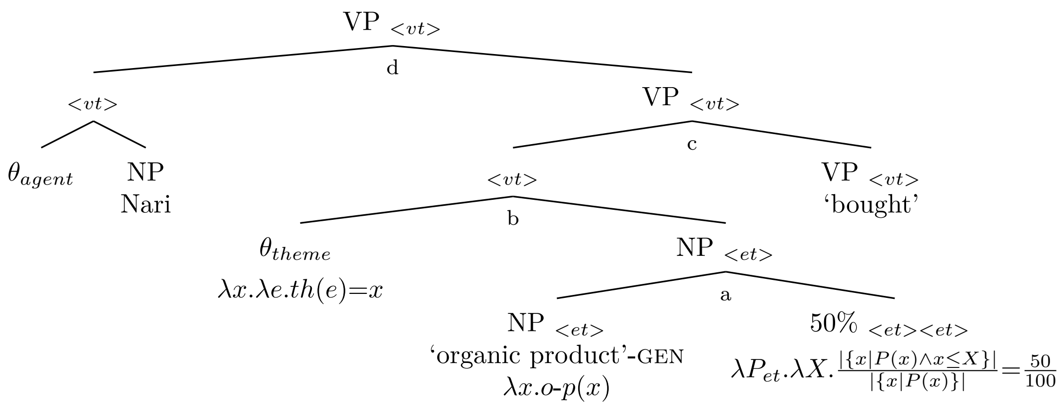
‘Nari bought 50% of the organic products.’ (✓proportional)

b. Nari-ka [yukinong ceyphum]-ul **50%** sassta
Nari-NOM organic product-ACC 50% bought

‘50% of what Nari bought was organic products.’ (✓reverse proportional)

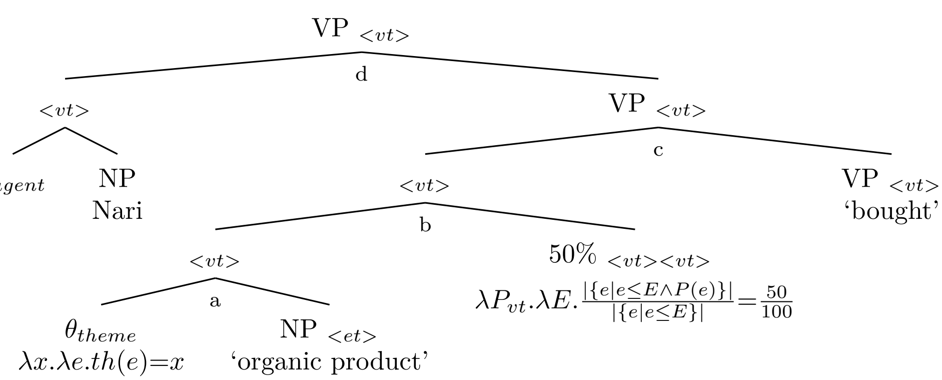
(83) and (84) show the derivations of the proportional sentence (82a) and the reverse proportional sentence (82b). Proportional and reverse proportional % have the same meaning as in subject-oriented sentences, but differ in that they target theme arguments. As noted in Section 4.2.1, event maximization is assumed, though it is not explicitly represented for simplicity.

(83) Proportional measure phrase



$[\![a]\!] = \lambda X.(\frac{|\{x|o\text{-}p(x)\wedge x\leq X|\}|}{|\{x|o\text{-}p(x)\}|} = \frac{50}{100})$
$[\![b]\!] = \lambda E.\exists X.th(E) = X \wedge (\frac{|\{x|o\text{-}p(x)\wedge x\leq X\}|}{|\{x|o\text{-}p(x)\}|} = \frac{50}{100})$ (via RESTRICT, $\exists_X$)
$[\![c]\!] = \lambda E.\exists X.buy(E) \wedge th(E) = X \wedge (\frac{|\{x|o\text{-}p(x)\wedge x\leq X\}|}{|\{x|o\text{-}p(x)\}|} = \frac{50}{100})$
$[\![d]\!] = \lambda E.\exists X.buy(E) \wedge ag(E) = nari \wedge th(E) = X \wedge (\frac{|\{x|o\text{-}p(x)\wedge x\leq X\}|}{|\{x|o\text{-}p(x)\}|} = \frac{50}{100})$
$\rightarrow \exists E.\exists X.buy(E) \wedge ag(E) = nari \wedge th(E) = X \wedge (\frac{|\{x|o\text{-}p(x)\wedge x\leq X\}|}{|\{x|o\text{-}p(x)\}|} = \frac{50}{100})$ (VIA $\exists_E$)

(84) Reverse proportional measure phrase



$[\![a]\!] = \lambda e.\exists x.th(e) = x \wedge o\text{-}p(x)$ (via RESTRICT$_E$)
$[\![b]\!] = \lambda E.\frac{|\{e|e\leq E\wedge\exists x.th(e)=x\wedge o\text{-}p(x)\}|}{|\{e|e\leq E\}|} = \frac{50}{100}$
$[\![c]\!] = \lambda E.buy(E) \wedge \frac{|\{e|e\leq E\wedge\exists x.th(e)=x\wedge o\text{-}p(x)\}|}{|\{e|e\leq E\}|} = \frac{50}{100}$
$[\![d]\!] = \lambda E.buy(E) \wedge ag(E) = nari \wedge \frac{|\{e|e\leq E\wedge\exists x.th(e)=x\wedge o\text{-}p(x)\}|}{|\{e|e\leq E\}|} = \frac{50}{100}$
$\rightarrow \exists E.buy(E) \wedge ag(E) = nari \wedge \frac{|\{e|e\leq E\wedge\exists x.th(e)=x\wedge o\text{-}p(x)\}|}{|\{e|e\leq E\}|} = \frac{50}{100}$ (via $\exists_E$)

As with the count nouns in (82), mass nouns can also serve as event participants, as shown in (85).

(85) a. Nari-ka [maykcwu-uy **50%**]-lul masyessta
Nari-NOM beer-GEN 50%-ACC drank

'Nari drank 50% of the beer.' (✓proportional)

b. Nari-ka maykcwu-lul **50%** masyessta
Nari-NOM beer-ACC 50% drank

'50% of what Nari drank was beer.' (✓reverse proportional)

English does not allow subject-oriented reverse sentences, but it does allow object-oriented reverse ones. The English object-oriented measurement sentences in (86) have the same structure and composition as their Korean counterparts in (83) and (84), differing only in their head-initial linearization.

(86) a. Nari bought **50%** of the organic products. (✓proportional)

b. Nari bought **50%** organic products.
'50% of what Nari bought was organic products.' (✓reverse proportional)

4.2.3 Goal

Measure phrases can be associated with theme and goal arguments in ditransitive constructions, realized as direct and indirect objects. In (87), the word order among internal arguments is flexible without affecting reverse proportional interpretation, with even greater variation when they precede the external argument. For the purposes of this discussion, I fix the external argument in sentence-initial position. The structures and derivations corresponding to the three interpretations in (87) are presented below, setting aside event maximization for simplicity. The denotation of % in theme-oriented measurement sentences from the previous subsection carries over straightforwardly.

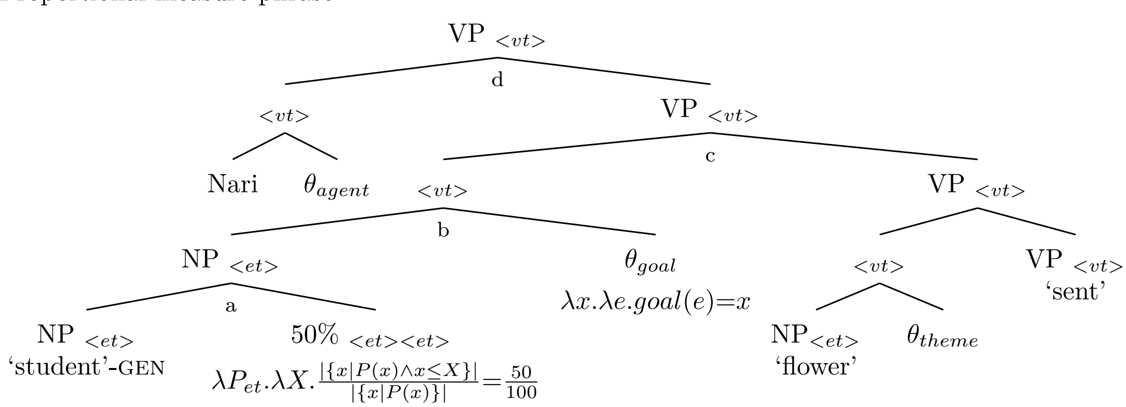
(87) a. Nari-ka {[haksayng-uy **50%**]-eykey} {kkoch-ul} ponayssta
Nari-NOM student-GEN 50%-DAT flower-ACC sent

'Nari sent flowers to 50% of the students.' (✓proportional on the goal)

b. Nari-ka {haksayng-eykey} {kkoch-ul} {**50%**} ponayssta
Nari-NOM student-DAT 50% flower-ACC sent

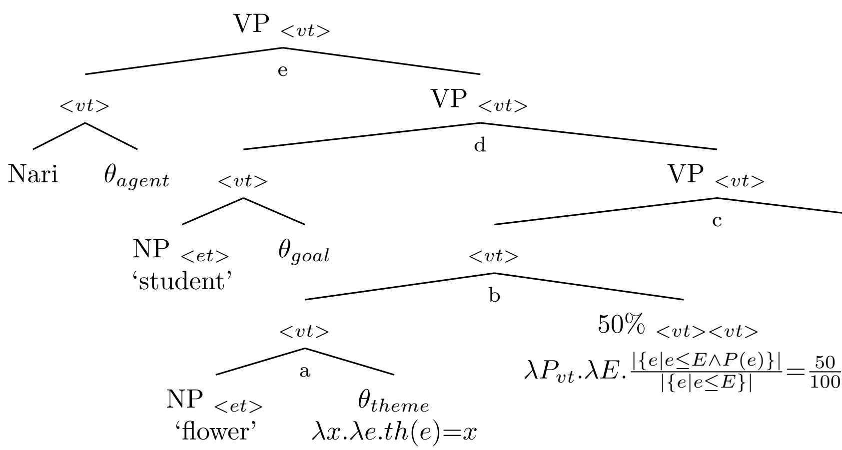
'50% of what Nari sent to the students were flowers.' (✓reverse proportional on the theme)
'50% of the people Nari sent flowers to were students.' (✓reverse proportional on the goal)

(88) Proportional measure phrase



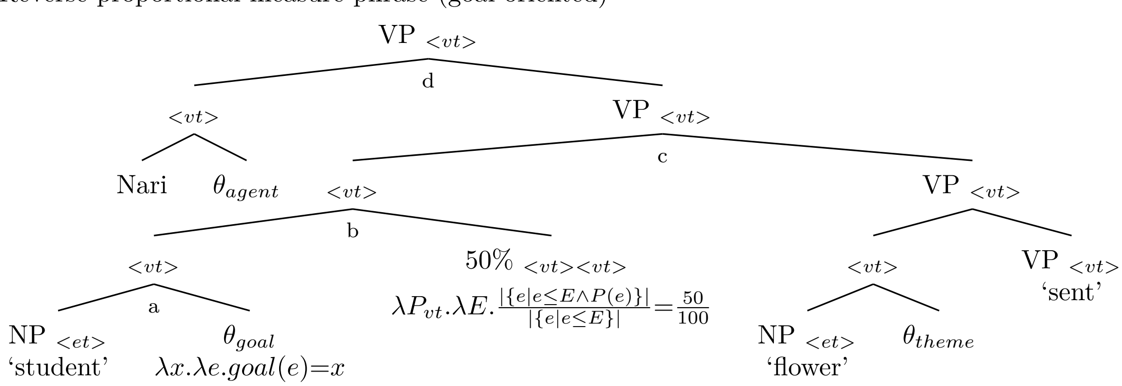
$[\![a]\!] = \lambda X.\frac{|\{x|student(x)\wedge x\leq X\}|}{|\{x|student(x)\}|} = \frac{50}{100}$
$[\![b]\!] = \lambda E.\exists X.goal(E) = X \wedge (\frac{|\{x|student(x)\wedge x\leq X\}|}{|\{x|student(x)\}|} = \frac{50}{100})$ (via RESTRICT, $\exists_X$)
$[\![c]\!] = \lambda E.\exists X.\exists Y.sent(E) \wedge goal(E){=}X \wedge th(E){=}Y \wedge flower(Y) \wedge (\frac{|\{x|student(x)\wedge x\leq X\}|}{|\{x|student(x)\}|}{=}\frac{50}{100})$
$[\![d]\!] = \lambda E.\exists X.\exists Y.sent(E){\wedge}ag(E){=}nari{\wedge}goal(E){=}X{\wedge}th(E){=}Y{\wedge}flower(Y){\wedge}(\frac{|\{x|student(x)\wedge x\leq X\}|}{|\{x|student(x)\}|}{=}\frac{50}{100})$
$\rightarrow \exists E.\exists X.\exists Y.sent(E){\wedge}ag(E){=}nari{\wedge}goal(E){=}X{\wedge}th(E){=}Y{\wedge}flower(Y){\wedge}(\frac{|\{x|student(x)\wedge x\leq X\}|}{|\{x|student(x)\}|}{=}\frac{50}{100})$

(89) Reverse proportional measure phrase (theme-oriented)



$[\![a]\!] = \lambda e.\exists x.th(e) = x \wedge flower(x)$ (via RESTRICT$_E$)
$[\![b]\!] = \lambda E.(\frac{|\{e\leq E|\exists x.th(e)=x\wedge flower(x)\}|}{|\{e|e\leq E\}|} = \frac{50}{100})$
$[\![c]\!] = \lambda E.sent(E) \wedge (\frac{|\{e\leq E|\exists x.th(e)=x\wedge flower(x)\}|}{|\{e|e\leq E\}|} = \frac{50}{100})$
$[\![d]\!] = \lambda E.\exists X.sent(E) \wedge goal(E){=}X \wedge student(X) \wedge (\frac{|\{e\leq E|\exists x.th(e)=x\wedge flower(x)\}|}{|\{e|e\leq E\}|}{=}\frac{50}{100})$
$[\![e]\!] = \lambda E.\exists X.sent(E){\wedge}ag(E){=}nari{\wedge}goal(E){=}X{\wedge}student(X){\wedge}(\frac{|\{e\leq E|\exists x.th(e)=x\wedge flower(x)\}|}{|\{e|e\leq E\}|}{=}\frac{50}{100})$
$\rightarrow \exists E.\exists X.sent(E){\wedge}ag(E){=}nari{\wedge}goal(E){=}X{\wedge}student(X){\wedge}(\frac{|\{e\leq E|\exists x.th(e)=x\wedge flower(x)\}|}{|\{e|e\leq E\}|}{=}\frac{50}{100})$

(90) Reverse proportional measure phrase (goal-oriented)



$[\![a]\!] = \lambda e.\exists x.goal(e) = x \wedge student(x)$ (via RESTRICT$_E$)
$[\![b]\!] = \lambda E.(\frac{|\{e\leq E|\exists x.goal(e)=x\wedge student(x)\}|}{|\{e|e\leq E\}|} = \frac{50}{100})$
$[\![c]\!] = \lambda E.\exists X.sent(E) \wedge th(E){=}X \wedge flower(X) \wedge (\frac{|\{e\leq E|\exists x.goal(e)=x\wedge student(x)\}|}{|\{e|e\leq E\}|}{=}\frac{50}{100})$
$[\![d]\!] = \lambda E.\exists X.sent(E){\wedge}ag(E){=}nari{\wedge}th(E){=}X{\wedge}flower(X){\wedge}(\frac{|\{e\leq E|\exists x.goal(e)=x\wedge student(x)\}|}{|\{e|e\leq E\}|}{=}\frac{50}{100})$
$\rightarrow \exists E.\exists X.sent(E){\wedge}ag(E){=}nari{\wedge}th(E){=}X{\wedge}flower(X){\wedge}(\frac{|\{e\leq E|\exists x.goal(e)=x\wedge student(x)\}|}{|\{e|e\leq E\}|}{=}\frac{50}{100})$

4.2.4 Time and location

(91) shows that proportional measure phrases can also target temporal and locative arguments.

(91) a. Podo-ka pam-ey **50%** cicessta
Podo-NOM night-at **50%** barked

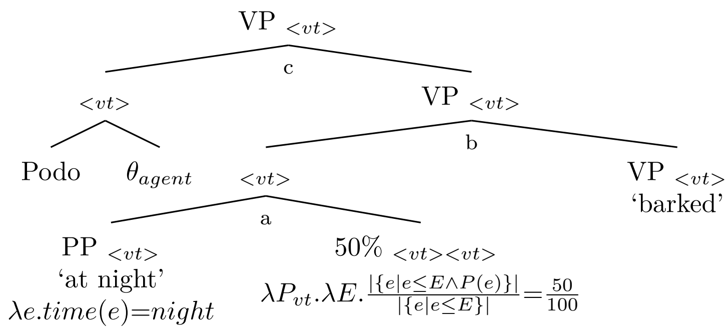
‘50% of Podo’s barking happened at night.’ (✓reverse proportional)

b. Podo-ka kongwen-eyse **50%** cicessta
Podo-NOM park-in **50%** barked

‘50% of Podo’s barking happened in the park.’ (✓reverse proportional)

The neo-Davidsonian approach enables a natural extension of event-related *%* to these cases, because canonical arguments and temporal/locative modifiers are both predicates of event. (92) provides the derivation of (91a). As with events involving a mass-noun theme, what counts as the atomic parts of the barking event is somewhat vague, though the event should still be divisible into contextually supported minimal parts.

(92)



$[\![a]\!] = \lambda E.\frac{|\{e|e\leq E\wedge time(e)=night\}|}{|\{e|e\leq E\}|} = \frac{50}{100}$
$[\![b]\!] = \lambda E.bark(E) \wedge \frac{|\{e|e\leq E\wedge time(e)=night\}|}{|\{e|e\leq E\}|} = \frac{50}{100}$
$[\![c]\!] = \lambda E.bark(E) \wedge ag(E) = podo \wedge \frac{|\{e|e\leq E\wedge time(e)=night\}|}{|\{e|e\leq E\}|} = \frac{50}{100}$
$\rightarrow \exists E.bark(E) \wedge ag(E) = podo \wedge \frac{|\{e|e\leq E\wedge time(e)=night\}|}{|\{e|e\leq E\}|} = \frac{50}{100}$ (via $\exists_E$)

4.3 Event-sensitive restrictions

This section examines whether the event-sensitive restrictions observed in other split quantification, specifically Japanese floating quantifiers and French quantification at a distance, also apply to reverse proportional measure phrases. I show how each restriction follows from the proposed semantics.

4.3.1 Collective interpretations

Collective interpretations are not allowed for *50%* in reverse proportional measurement sentences. While (93a) allows both distributive and collective proportional readings, (93b) does not have a collective reverse proportional reading.

(93) a. [penguin-uy **50%**]-ka tungci-lul ciessta
penguin-GEN 50%-NOM nest-ACC built

‘50% of the penguins built a nest.’
(✓distributive proportional; ✓collective proportional)

b. penguin-i **50%** tungci-lul ciessta
penguin-NOM 50% nest-ACC built

‘50% of those who built a nest were penguins.’
(✓distributive reverse proportional; ✗collective reverse proportional)

The lack of collective interpretations in (93b) follows from the non-conservative $\%_{event}$, repeated in (94a). The simplified truth conditions of (93b) are shown in (94b).

(94) a. $[\![\%_{event.noncon}]\!] = \lambda d.\lambda P.\lambda E.\frac{|\{e|P(e)\wedge e\leq E\}|}{|\{e|e\leq E\}|} = \frac{d}{100}$

b. $[\![(93b)]\!] = \exists E.built(E) \wedge th(E) = nest \wedge \frac{|\{e|\exists x.ag(e)=x\wedge penguin(x)\wedge e\leq E\}|}{|\{e|e\leq E\}|} = \frac{50}{100}$

Collective events constitute a single event with a single group agent (Landman 2000). In collective contexts, therefore, the denominator in (94b) is always 1, and the numerator can only be 1 as well, given the atomic group agent. The formula is a contradiction, and the collective reverse interpretation is correctly ruled out. Note that if the event E has collective events e as proper parts, the interpretation is no longer truly collective. Each collective event is treated as an atomic subevent, which essentially amounts to a distributive interpretation. The reverse proportional reading of (93b) is acceptable only under such distributive scenarios.

One might then wonder whether 100% licenses reverse proportional readings in collective sentences, specifically, when a single atomic group of penguins is the agent of the collective nest-building event and $d = 100$. (95) shows that this is indeed the case. In a context with many penguins and ducks, (95b) is felicitous when describing a single nest collectively built by five penguins.

(95) a.

[penguin-uy	**100%**]-ka	tungci-lul	ciessta
penguin-GEN	100%-NOM	nest-ACC	built

'100% of the penguins built a nest.'
(✓distributive proportional; ✓collective proportional)

b.

penguin-i	**100%**	tungci-lul	ciessta
penguin-NOM	100%	nest-ACC	built

'100% of those who built a nest were penguins.'
(✓distributive reverse proportional; ✓collective reverse proportional)

Turning to the object-oriented sentences, collective interpretations are blocked in (96b) when 50% is intended as reverse proportional. Consider a context in which Nari, an entrepreneur with several for-profit companies and non-profit organizations, decides to carry out a large-scale merger. Again, I set aside meta-collective readings, which are essentially distributive and attested.

(96) a.

Nari-ka	[piyengli	tanchey-uy	**50%**]-lul	happyenghayssta
Nari-NOM	nonprofit	organization-GEN	50%-ACC	merged

'Nari merged 50% of the non-profits.' (✓proportional)

b.

Nari-ka	piyengli	tanchey-lul	**50%**	happyenghayssta
Nari-NOM	nonprofit	organization-ACC	50%	merged

'50% of what Nari merged were non-profits.' (✗reverse proportional)

The truth conditions of (96b) are shown in (97).

(97) $[\![(96b)]\!] = \exists E.merge(E) \wedge ag(E) = nari \wedge \frac{|\{e|e\leq E\wedge\exists x.th(e)=x\wedge non\text{-}profit(x)\}|}{|\{e|e\leq E\}|} = \frac{50}{100}$

We are dealing with a collective event in which Nari merged some of her for-profit and non-profit organizations into a single entity. Assuming that collective events are atomic, E has only one subevent; thus, the sentence in (96b) is a contradiction. The only collective scenario where the truth conditions can be satisfied is when all merged entities are non-profits; accordingly, the theme is an atomic group of non-profits and $d = 100$. The sentence with *100%* is acceptable, as illustrated in (98b).

(98) a. Nari-ka [piyengli tanchey-uy **100%**]-lul happyenghayssta
Nari-NOM nonprofit organization-GEN 50%-ACC merged

'Nari merged 100% of the non-profits.' (✓proportional)

b. Nari-ka piyengli tanchey-lul **100%** happyenghayssta
Nari-NOM nonprofit organization-ACC 100% merged

'100% of what Nari merged were non-profits.' (✓reverse proportional)

4.3.2 Statives

Second, Nakanishi (2007) showed that individual-level predicates are incompatible with Japanese numerals and measure phrases in split quantification. (99) demonstrates that individual-level predicates do not support reverse proportional readings of *50%* in Korean split quantification, unless they are coerced into eventive progressives, as in 'are acting aggressively.'

(99) a. [hama-uy **50%**]-ka kongkyekcekita
hippo-GEN 50%-NOM aggressive

'50% of the hippos are aggressive.' (✓proportional)

b. hama-ka **50%** kongkyekcekita
hippo-NOM 50% aggressive

'50% of the aggresive ones are hippos.' (✗reverse proportional)

Japanese floating quantifiers in split quantification could occur with stage-level predicates. This is not the case for proportional measure phrases. (100) shows that reverse proportional readings are not allowed even with stage-level predicates.

(100) a. [hama-uy **50%**]-ka kenkanghata
hippo-GEN 50%-NOM healthy

'50% of the hippos are healthy.' (✓proportional)

b. hama-ka **50%** kenkanghata
hippo-NOM 50% healthy

'50% of the healthy ones are hippos.' (✗reverse proportional)

Given (99) and (100), the generalization is that reverse proportional measure phrases are not compatible with either type of adjectival statives. Verbal statives, as in (101b), also resist reverse proportional readings unless aspectually coerced. The reading improves only under aspectual coercion when *like* is reinterpreted inchoatively. This is particularly relevant for statives in Korean, where judgments can be delicate: stative verbs like *know*, *have*, *like* allow progressive morphology and can be interpreted as inchoatives (Lee 2006).

(101) a. Nari-ka [koyangi-uy **50%**]-lul cohahanta
Nari-GEN cat-GEN 50%-ACC like

'Nari likes 50% of the cats.' (✓proportional)

b. Nari-ka koyangi-lul **50%** cohahanta
Nari-NOM cat-ACC 50% like

'50% of what Nari likes is cats.' (✗reverse proportional)

Stative predicates fail to generate reverse proportional readings because, as explained in Section 3.3.2, states cannot be discretely individuated and counted in the way events can be. As a result, the denominator in (102b) is undefined; consequently, the (b) sentences in (99-101) are correctly predicted to be unacceptable.

(102) a. $[\![(99a)]\!] = \exists S.\exists X.holder(S) = X \wedge aggressive(S) \wedge \frac{|\{x|hippo(x) \wedge x \leq X\}|}{|\{x|hippo(x)\}|} = \frac{50}{100}$

b. $[\![(99b)]\!] = \exists S.aggressive(S) \wedge \frac{|\{s|s \leq S \wedge \exists x.holder(s) = x \wedge hippo(x)\}|}{|\{s|s \leq S\}|} = \frac{50}{100}$

To sum up, apart from eventive coercion, both adjectival and verbal statives in reverse proportional sentences are unacceptable because states on their own are incompatible with the part-whole-based individuation and counting. Crucially, this also explains why reverse proportional % must quantify over events rather than individuals. An individual-based % would incorrectly predict split-quantification sentences with statives to be acceptable. For example, under the purely individual-based quantification, there is no issue with counting the atomic parts x of a plurality of hippos X that are healthy, yielding a reverse proportional interpretation in which they make up 50% of the total healthy individuals.

4.3.3 Single events

The third property of split configurations is that they are incompatible with single-occurrence events. In the context of proportional quantification, single-occurrence events should involve *100%* to begin with. I first establish the acceptable baseline with a minimal pair involving *100%*.

(103) a. [penguin-uy **100%**]-ka talchulhayssta
penguin-GEN 100%-NOM escaped

'100% of the penguins escaped.' (✓proportional)

b. penguin-i **100%** talchulhayssta
penguin-NOM 100% escaped

'100% of the escapees were penguins.' (✓reverse proportional)

To test single-occurrence events, we can distinguish two contexts: collective single events and non-collective single events. The former was already discussed above, so in what follows I focus on the latter. While the sentence in (103b) is perfectly acceptable in general, if there was only one escaping event and it involved a penguin, the sentence becomes marginal. However, this effect is not specific to reverse proportional measure phrases: (103a) is equally marginal in a context where there was only one penguin and it escaped. This parallel is illustrated in (104).

(104) a. [penguin-uy **100%**]-ka talchulhayssta
penguin-GEN 100%-NOM escaped

'100% of the penguins escaped.' (?? if there was only one penguin)

b. penguin-i **100%** talchulhayssta
penguin-NOM 100% escaped

'100% of the escapees were penguins.' (?? if there was only one escaping)

Since both (104a) and (104b) show the same degree of marginality, I propose that the degraded acceptability with single-event contexts arises from the trivially small quantification domain. The present account of *%* does in fact permit singleton cases. For example, if there was exactly one penguin, the sentence *100% of the penguins escaped* is truth-conditionally satisfied if that penguin escaped. Nevertheless, such sentences feel odd in singleton contexts because the domain is trivially small.

To see how the pragmatic markedness arises from the trivially small domain, in singleton contexts, simply stating 'penguin-NOM escaped' without *100%* conveys the same information. That is, the simpler alternative without *100%* is equally informative as the literal meaning of (104a). Using a more complex construction with *100%* in such contexts signals that the intended meaning differs from the simpler alternative, assuming the speaker is cooperative. However, in singleton contexts, *100%* adds no extra meaning; thus, the speaker is effectively violating the cooperative principle by using an unnecessarily complex form, leading to the degraded judgment. This pattern parallels other quantificational cases, such as *Every student came*, which is truth-conditionally satisfied when there is only one student, but pragmatically marked.

The same pattern extends to theme-oriented measurement sentences such as (105). While these sentences form an acceptable minimal pair in most contexts, they become degraded when the relevant domain is a singleton: for (105a), when there was only one organic product, and for (105b), when Nari bought only one item and it was an organic product.

(105) a. Nari-ka [yukinong ceyphum-uy **100%**]-lul sassta
Nari-NOM organic product-GEN 100%-ACC bought

'Nari bought 100% of the organic products.' (?? if there was only one organic product)

b. Nari-ka yukinong ceyphum-ul **100%** sassta
Nari-NOM organic product-ACC 100% bought

'100% of what Nari bought was organic products.' (?? if Nari bought only one item)

4.4 Multiple case constructions

This section addresses an additional complexity in split quantification constructions that has been set aside until now. Split quantification examples, often the (b) sentences in minimal pairs, have been shown to allow reverse proportional readings, analyzed as subevent quantification. However, in Korean, measure phrases like *50%* in split configurations can also be interpreted proportionally, unlike in English. This is illustrated in (106) and (107), where the examples are repeated from Section 4.2, but with an additional available interpretation for the (b) sentences.

(106) a. [hoywen-uy **50%**]-ka censi-lul kwanlamhayssta
member-GEN 50%-NOM exhibition-ACC viewed

'50% of the members viewed the exhibition.' (✓proportional)

b. hoywen-i **50%** censi-lul kwanlamhayssta
member-NOM 50% exhibition-ACC viewed

'50% of the members viewed the exhibition.' (✓proportional)
'50% of the exhibition's viewers were members.' (✓reverse proportional)

(107) a. Nari-ka [yukinong ceyphum-uy **50%**]-lul sassta
Nari-NOM organic product-GEN 50%-ACC bought

'Nari bought 50% of the organic products.' (✓proportional)

b. Nari-ka yukinong ceyphum-ul **50%** sassta
Nari-NOM organic product-ACC 50% bought

'Nari bought 50% of the organic products.' (✓proportional)
'50% of what Nari bought was organic products.' (✓reverse proportional)

Given the proposed analysis of *%*, the remaining question is how to account for the additional proportional reading in split quantification sentences. I argue that these proportional interpretations arise from the Multiple Nominative Constructions (MNC) and Multiple Accusative Constructions (MAC), a productive case-marking pattern in Korean. The idea is that when sentences with a split measure phrase are interpreted reverse proportionally, *50%* is implicitly marked nominative or accusative, an option also suggested by Ahn and Ko (2022) as a possible source of ambiguity. Therefore, under the proportional interpretation, a second case marker is present in the structure but is unpronounced. Three empirical facts support this connection between multiple case constructions and proportional readings in split sentences. First, explicitly case-marked MNC (108) and MAC (109) sentences allow only proportional readings.

(108) hoywen-i **50%**-ka censi-lul kwanlamhayssta
member-NOM 50%-NOM exhibition-ACC viewed

'50% of the members viewed the exhibition.' (✓proportional)
'50% of the exhibition's viewers were members.' (✗reverse proportional)

(109) Nari-ka yukinong ceyphum-ul **50%**-lul sassta
Nari-NOM organic product-ACC 50%-ACC bought

'Nari bought 50% of the organic products.' (✓proportional)
'50% of what Nari bought was organic products.' (✗reverse proportional)

Second, although multiple nominative and accusative constructions are a well-documented property of Korean and Japanese (e.g., Youn (1990), Heycock (1993)), multiple case constructions do not occur with datives in Korean. Against this background, (110b) does not show ambiguity; it cannot be interpreted proportionally with respect to the dative argument. The lack of proportional readings for datives strongly suggests that the additional proportional interpretations in split configurations (45) and (46) arise due to multiple nominative and accusative constructions.

(110) a. Nari-ka [haksayng-uy **50%**]-eykey kkoch-ul ponayssta
Nari-NOM student-GEN 50%-DAT flower-ACC sent

'Nari sent flowers to 50% of the students.' (✓proportional)

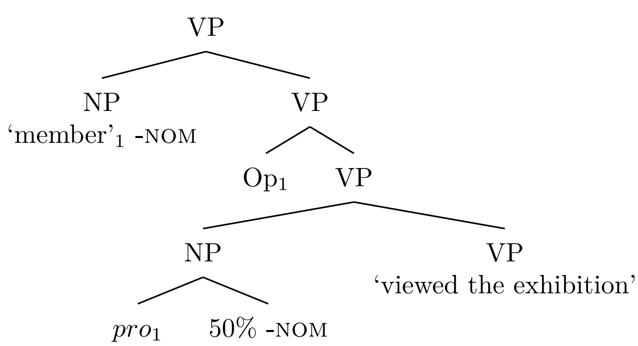
b. Nari-ka haksayng-eykey **50%** kkoch-ul ponayssta
Nari-NOM student-DAT 50% flower-ACC sent

'Nari sent flowers to 50% of the students.' (✗proportional)
'50% of the flowers Nari sent were to students.' (✓reverse proportional)

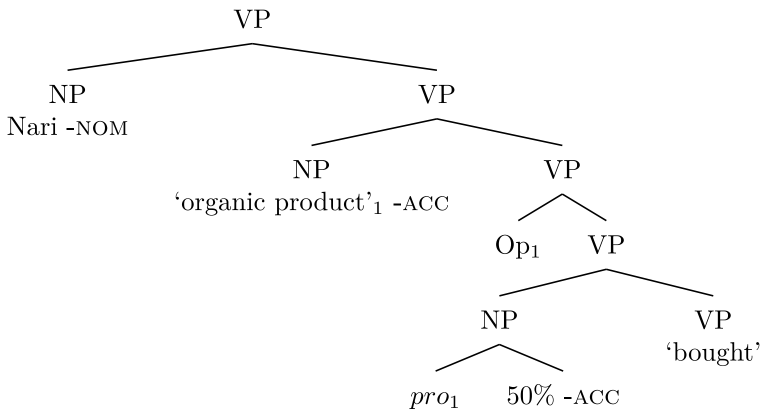
Last but not least, as with pro-drop, case markers in Korean are frequently dropped, especially in spoken language. I leave open the precise empirical generalizations and the mechanisms governing when and how case-drop occur, as these questions are beyond the scope of this work.

Setting aside the detailed analysis of multiple case constructions, I assume that the additional predications in MNC and MAC are mediated by a null operator, in the spirit of Heycock (1993), who argues that multiple layers of predication can coexist in a single sentence. In these multiply case-marked constructions, *50%* is a standard nominal modifier over the individual domain, generating proportional readings. For simplicity, thematic predicates are ignored in (111) and (112).

(111) Multiple Nominative Construction



(112) Multiple Accusative Construction



In summary, split measure phrases in (106b) and (107b) can receive both proportional and reverse proportional interpretations. Under proportional readings, these sentences are multiply case-marked constructions, as in (108) and (109), where *50%* is implicitly case-marked and functions as a modifier in the individual domain, with structures illustrated in (111) and (112). By contrast, under reverse proportional readings, *50%* is a modifier in the event domain, as proposed in Section 3.2.

4.5 Attributive uses

The measure phrase *50%* can be used attributively, as shown by Coppock (2022) in (113).

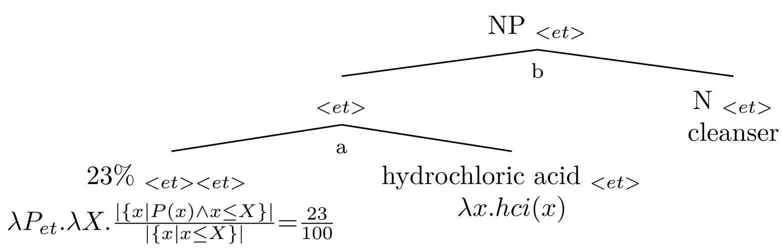
(113) Attributive *%*

a. I found [a **23%** hydrochloric acid cleanser].

b. The cleanser is **23%** hydrochloric acid.

I analyze the measure phrases in (113) as having non-conservative, individual-based denotations, as their second arguments are clearly individual-based (i.e., (the) cleanser), regardless of the syntactic category of the first argument. The structure and composition of the NP object in (113a) are as follows:

(114)



$[\![a]\!] = \lambda X.\frac{|\{x|hci(x)\wedge x\leq X\}|}{|\{x|x\leq X\}|} = \frac{23}{100}$
$[\![b]\!] = \lambda X.cleanser(X) \wedge \frac{|\{x|hci(x)\wedge x\leq X\}|}{|\{x|x\leq X\}|} = \frac{23}{100}$ (⇒ type-shifting via IOTA or A)

Although what counts as an atomic part of a cleanser may be less straightforward than for plural nouns, I assume that x refers to contextually supported minimal parts of the cleanser X, which aligns with the primary function of measure nouns (as with classifiers) in relating numbers to objects. For concreteness, I continue to use discrete cardinality functions, though switching to continuous measure functions remains an analytical option if needed.

Given (114), the sentence in (113b) is simply a complex attributive use such as (113a) turned into a small clause (specifically, a copular clause), with the second argument X now serving as the subject (Coppock reports that in addition to copulas, verbs like *remain*, *become*, *consist of* can also appear in these constructions, making reference to subparts of the subject). To examine this type of copular clause with an individual-denoting subject in detail, consider the following minimal pair in Korean.

(115) a. [supak-uy **92%**]-ka mulita
watermelon-GEN 92%-NOM water.COP

‘92% of watermelon is water.’ (✓proportional)

b. supak-i **92%** mulita
watermelon-NOM 92% water.COP

‘92% of watermelon is water.’ (✓)
‘92% of water is watermelon.’ (✗)

I did not characterize each reading in (115b), since in Korean there are, in principle, two possible ways to describe the available reading in the first line. On the one hand, the available reading in (115b) can be pre-analytically described as a proportional reading, given the discussion of multiple nominative constructions in Section 4.4. Indeed, a string with a second overt nominative on 92% sounds as natural as (115b) and conveys exactly the same meaning as (115a). On the other hand, 92% in (115b) can instead be associated with the NP predicate. In this case, we could reasonably characterize the available reading as reverse proportional, though not event-based, since the second argument (*watermelon* in (115b) or *the cleanser* in (113b)) clearly denotes individuals.

There is a reason to favor the latter analysis, in which 92% in (115b) is associated with the predicate rather than the subject. English lacks multiple nominative constructions, but (116), like (113b), is perfectly acceptable with the meaning that water makes up 92% of watermelon, similar to Korean (115b). A crosslinguistic parallel supports analyzing 92% in both (116) and (115b) as forming a constituent with the predicate.

(116) Watermelon is 92% water.
‘92% of watermelon is water.’ (✓)
‘92% of water is watermelon.’ (✗)

As previewed above, I analyze (115b) and (116) (as well as (113b)) as attributive uses converted into small-clause contexts. (117) illustrates the derivation. The subject may be either definite or indefinite; here I assume an indefinite argument that is existentially closed. The measure phrase 92% in (117) is an individual-quantifying modifier with a non-conservative denotation, just like 92% in its attributive uses in (113a) and (114).

(117) PredP $_{t}$ (b)
- NP $_{<et>}$ watermelon
- PredP $_{<et>}$ (a)
  - 92% $_{<et><et>}$ $\lambda P_{et}.\lambda X.\frac{|\{x|P(x)\wedge x\leq X\}|}{|\{x|x\leq X\}|}=\frac{92}{100}$
  - Pred $_{<et>}$ water $\lambda x.water(x)$

$[\![a]\!] = \lambda X.\frac{|\{x|water(x)\wedge x\leq X\}|}{|\{x|x\leq X\}|} = \frac{92}{100}$
$[\![b]\!] = \exists X.watermelon(X) \wedge \frac{|\{x|water(x)\wedge x\leq X\}|}{|\{x|x\leq X\}|} = \frac{92}{100}$

A potential complication arises from (116)'s resemblance to quantifier floating, a phenomenon also attested in English, as shown in (118):

(118) a. [**All** my friends] are artists.

b. My friends are [**all** ___] artists.

The difference between *all* in (118) and *92%* in (119) is that the partitive marker disappears in (119b) when the NP surfaces at the beginning of the sentence. This behavior is not specific to *%* but reflects a more general property of partitive *of* as a construction marker, as it can also appear with other quantifiers like *all* and *each*.

(119) a. [**92%** of watermelon] is water

b. Watermelon is [**92%** ___] water.

As introduced in Section 2.1, there are two main proposals for the syntax of floating quantifiers: the stranding analysis (Sportiche 1988; Shlonsky 1991; Merchant 1996; McCloskey 2000) and the adverbial analysis (Doetjes 1992; Baltin 1995; Torrego 1996; Bobaljik et al. 2003). Regardless of which syntactic analysis is correct for a given language, *92%* in (119b) can be interpreted as in (117) if it has to be interpreted next to the predicative NP. By contrast, if there is an independent reason for *92%* in (119b) to be interpreted next to the subject NP, as in (119a), the partitive, proportional meaning is always available. For those who argue for the latter, the tricky issue is the obligatory presence of partitive *of* in (119a), but not in (118b), which has indeed been cited as evidence against the stranding analysis in some adverbial accounts of floating quantifiers.

For the present purpose, the upshot is that even if these constructions are instances of quantifier floating, the proposed semantics of *%* applies straightforwardly once the appropriate syntactic analysis is established for the given language.

4.6 Section summary

This section has presented a compositional analysis of reverse proportional measure phrases grounded in subevent quantification. The analysis relies on two key components: part-whole-based polymorphic functions that apply in both the individual and event domains, and the structural position of arguments relative to thematic predicates that connect these domains. I have shown that proportional measure phrases can target different thematic relations in a principled way, extending naturally beyond core cases such as agents and themes to include goal arguments as well as temporal and locative adverbials. I also demonstrated how event-sensitive restrictions in canonical reverse measurement sentences follow from the proposed analysis.

A further contribution is an explanation of the ambiguity found in some split configurations in Korean, but not in English, where measure phrases can receive both proportional and reverse proportional interpretations. I argued that the additional proportional readings arise from implicit multiple case constructions, a productive case-marking pattern available in Korean but not in English. The section also discussed attributive uses of measure phrases, showing that measure phrases maintain the same meaning across the two different structural environments.

For both event-based and individual-based reverse proportional %, I have argued that the non-conservative denotation is consistently available in both Korean and English. One might object to the proposed analysis on theoretical grounds, namely, that it abandons strict conservativity for elements traditionally described as determiner-like. However, two theoretical considerations address this concern. First, Zuber and Keenan (2019) argue that weak conservativity, a principled relaxation of the classical Conservativity Constraint, remains restrictive enough to rule out nearly all unattested determiner meanings while still allowing non-conservative denotations. Second, previous analyses of reverse proportional measure phrases that preserve strict conservativity do so at a different theoretical cost. Ahn and Sauerland (2017), for instance, require a modified trace conversion rule proposed specifically for reverse proportional measure phrases, while Pasternak and Sauerland (2022) posit two distinct null operators to derive the correct truth conditions and resolve type mismatches. Taken together, these considerations suggest that strict conservativity alone should not outweigh other factors when evaluating competing analyses.

5 Conclusion

This paper has investigated the semantics of split quantification, focusing on three phenomena: floating quantifiers, quantification at a distance, and reverse proportional measure phrases. The central claim is that split quantification involves quantification over events rather than individuals, with two core compositional ingredients. First, split quantifiers are polymorphic: they denote the same function based on part-whole structure, applying uniformly to properties of individuals and properties of events. Second, the structural position of NP arguments relative to thematic predicates in a neo-Davidsonian framework correlates with the type of quantification: when the NP combines with the quantifier before the thematic predicate applies, individual quantification results; when the split quantifier applies after the NP combines with the thematic predicate, event quantification emerges.

For French quantification at a distance, I showed that while *beaucoup* 'a lot' in split configurations requires multiple events, as previously observed in the literature, the apparent multiplicity of objects arises pragmatically rather than being semantically encoded. For reverse proportional measure phrases such as *50%*, I demonstrated that a unified compositional template applies across various argument types as well as attributive uses. I also explained the interpretive ambiguity found in Korean split quantification sentences but not in English by appealing to multiple case constructions unique to Korean. The event-sensitive restrictions observed across these phenomena follow from the event-quantificational analysis. Overall, the proposed analysis offers a framework for further crosslinguistic investigation of other event-quantificational phenomena that surface morphosyntactically as split quantification.

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